



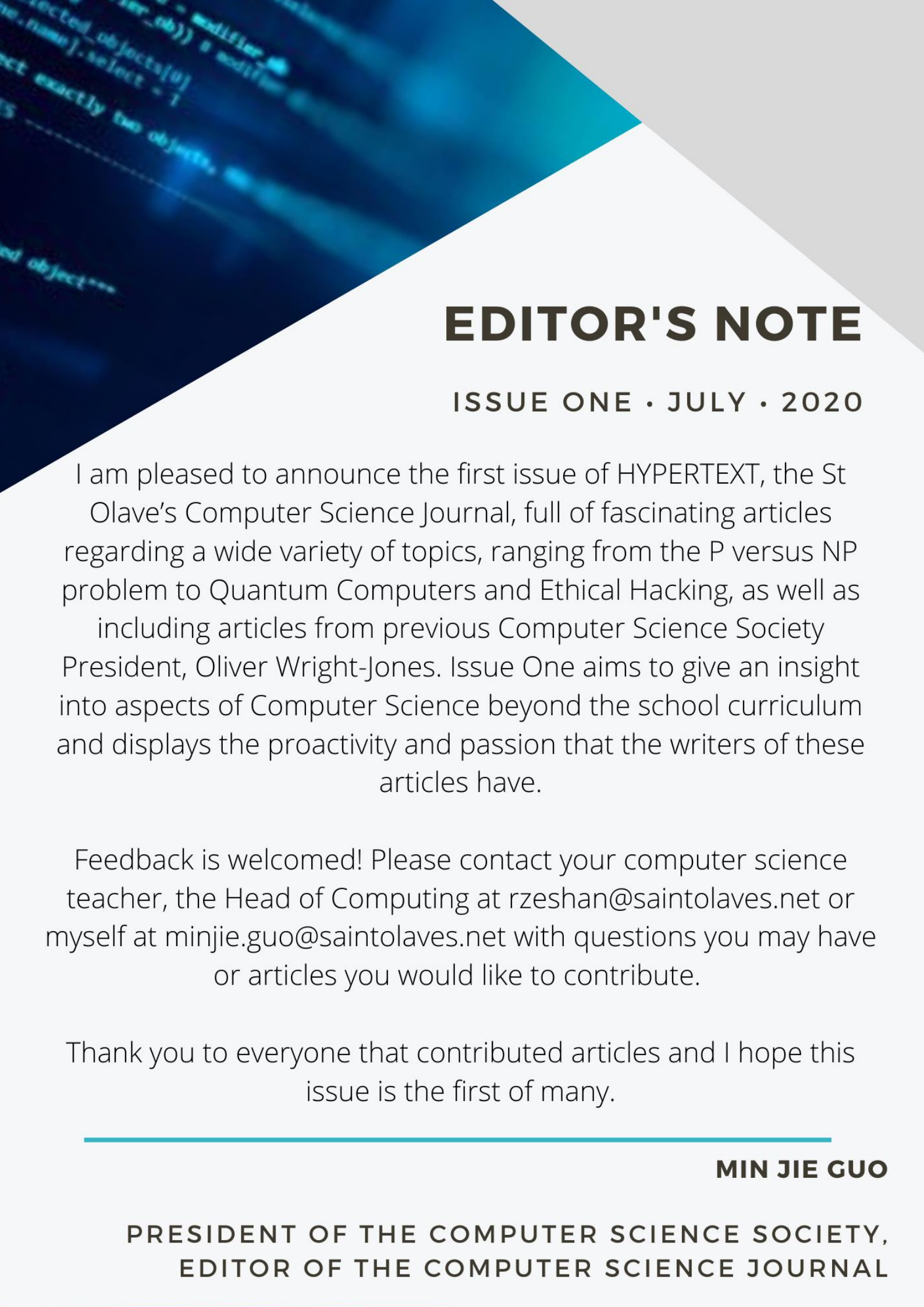
COMPUTER
SCIENCE SOCIETY

ST OLAVE'S
GRAMMAR SCHOOL

HYPertext

ISSUE ONE • JULY • 2020





EDITOR'S NOTE

ISSUE ONE • JULY • 2020

I am pleased to announce the first issue of HYPERTEXT, the St Olave's Computer Science Journal, full of fascinating articles regarding a wide variety of topics, ranging from the P versus NP problem to Quantum Computers and Ethical Hacking, as well as including articles from previous Computer Science Society President, Oliver Wright-Jones. Issue One aims to give an insight into aspects of Computer Science beyond the school curriculum and displays the proactivity and passion that the writers of these articles have.

Feedback is welcomed! Please contact your computer science teacher, the Head of Computing at rzeshan@saintolaves.net or myself at minjie.guo@saintolaves.net with questions you may have or articles you would like to contribute.

Thank you to everyone that contributed articles and I hope this issue is the first of many.

MIN JIE GUO

**PRESIDENT OF THE COMPUTER SCIENCE SOCIETY,
EDITOR OF THE COMPUTER SCIENCE JOURNAL**

CONTENTS

P=?NP

MIN JIE GUO AND
JANET THOMAS ALBERT PAGE 3

THE MATH BEHIND NEURAL
NETWORKS

ADVAIT MENON PAGE 6

ARE QUANTUM COMPUTERS AN
OPPORTUNITY OR THREAT?

RISHI TANDON PAGE 9

ENCRYPTION: THE QUANTUM
MECHANICS OF COMMUNICATION

LUCY STOCK PAGE 12

ALLIED CRYPTOSYSTEMS IN THE
SECOND WORLD WAR

OLIVER WRIGHT-JONES PAGE 18

TO WHAT EXTENT HAS DIGITAL
RIGHTS MANAGEMENT BEEN
EFFECTIVE IN PREVENTING
COPYRIGHT ABUSE?

OLIVER WRIGHT-JONES PAGE 21

ETHICAL HACKING

JAIPREET BAINS PAGE 26

SMART HOMES AND NETWORK
VULNERABILITIES: A RESIDENCE OF
WONDER OR RISK?

JANET THOMAS ALBERT PAGE 29

PASSWORDS AND
PASSWORD ENTROPY

THEVINDI PAGE 33
DODAMPEGAMAGE

NASH EQUILIBRIUM AND THE
PRISONER'S DILEMMA - GAME
THEORY

ADITYA PALANIAPPAN PAGE 35

ADDING INTELLIGENCE TO VIDEO
GAMES

ADVAIT MENON PAGE 37

AI IN MEDICINE

EKAKSH LAL PAGE 40

TRACKING THE SPREAD OF
CONTACT TRACING

PIERCE WIEGERLING PAGE 43

PYTHON COURSES - SHOULD
YOU DO THEM?

NEIL TRIPATHI PAGE 45

THE COMPUTER SCIENCE SOCIETY

MIN JIE GUO PAGE 48

P=?NP

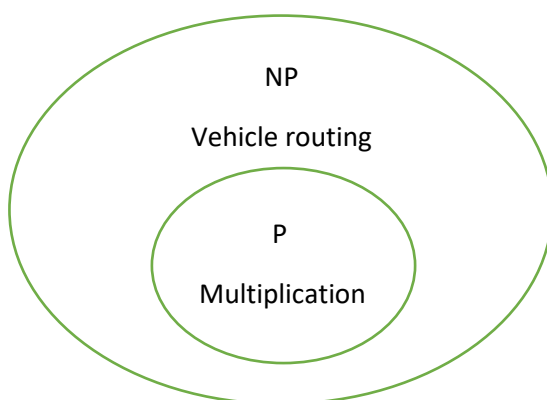
Min Jie Guo and Janet Thomas Albert

The philosophy of computer science is an engrossing area of study related to vital philosophical questions. One of the most gripping problems to arise from this field of study, quoted to be the 'most notorious problem in theoretical computer science', is the P versus NP problem, which still possesses no solution. It is one of the seven Millennium Prize Problems selected by the Clay Mathematics Institute, each of which carries a US\$1,000,000 prize for an acceptable answer proof.

After the invention of computers, programmers started writing the very first programs; these for the most part were adaptable and could be improved to increase efficiency leading to a faster run-time. But not all programs were as adaptable due to it being extremely difficult to find shorter solutions for very complicated programs. Therefore, people started organising programs in order of speed.

e.g.

FASTEST	SLOWEST
A multiplication program	Vehicle routing - 'What is the optimal set of routes for a fleet of vehicles to traverse to deliver to a given set of customers?'



A problem has many possible solutions and takes a certain number of inputs. Each potential solution may or may not solve the

problem for the given inputs. A problem is in NP (non-deterministic polynomial time) if it is possible to efficiently (so that an acceptable amount of time and space is taken) check a potential solution to see if it works. It is in P (polynomial time – the complexity of the algorithm is $O(n^k)$, where n = size of data, k = constant) if it is possible to efficiently find a solution, provided one exists.

P is the class which contains all the reasonably fast programs, whilst NP contains problems which are solved at a much slower rate. However, if a possible solution is given to a NP problem this can be computed to check whether it is correct - in a reasonable amount of time.

In some cases, problems first classified as NP were eventually re-classified as P as they were discovered to have a much simpler algorithm than first preconceived.

Hence, this beckons us to three critical questions:

Are the problems we have classified as NP all P problems? Is the split between NP and P problems truly valid?

Additionally, due to the prior mentioned characteristics of P and NP: does the ability to recognize the correct solution equate to there being a quicker way to solve the problem?

If this was the case, this would allude to $P=NP$ being true.

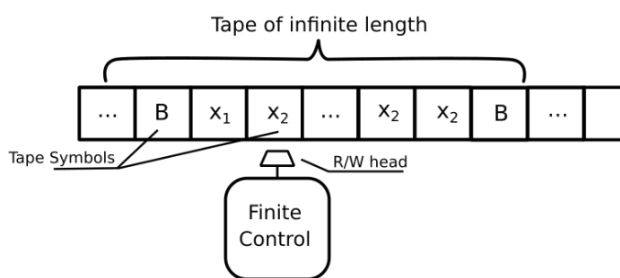
P can be written as $S = f(N)$, S being the number of steps, N being the size and f being a polynomial function, the vital idea being that it is much slower than an exponential function. NP on the other hand, simply put, means that with the power of an extremely large number of supercomputers which would check all possible solutions to your problem, you could find the solution in polynomial time.

The prospect of $P = NP$ would mean that many of the problems which we have been struggling to solve are easy for computers to

solve. It is also interesting to note that there are many other ways in which a problem can be classified apart from P and NP.

Neither conjecture of either $P=NP$ or $P \neq NP$ has been proven correct and this naturally irritates many mathematicians and computer scientists. This problem has been open since the early 1970s, so why has it not yet been solved?

The P versus NP problem is often modelled using a Turing Machine (TM), an abstract computational model, i.e. an imaginary computer. It has a finite number of states and an infinite tape, with discrete cells into which a limited set of symbols can be written to and read from. At any given time, the TM is in one of its states and is looking at a particular cell on the tape. Depending on what is read from the cell, it can write a new symbol into the cell, move the tape one cell forward or backward or go into a different state. This is called a state transition. A TM can be designed by a structure of different states and transitions and is equivalent to any computer program and therefore is used as a theoretical model for proving things about what computers can and cannot do.



A deterministic TM has only one transition from each state for each symbol that it is reading off the tape. A non-deterministic TM may have several transitions and hence can simultaneously check several possibilities, so can spawn as many threads (of execution, so a program can be split into two or more simultaneously running tasks) as needed. However, a real computer can only execute a specific number of threads at a time, equal to the number of CPUs (cores). Therefore, a non-deterministic TM is not seen as physically possible,

except perhaps with the use of a quantum computer with a significantly large number of cores.

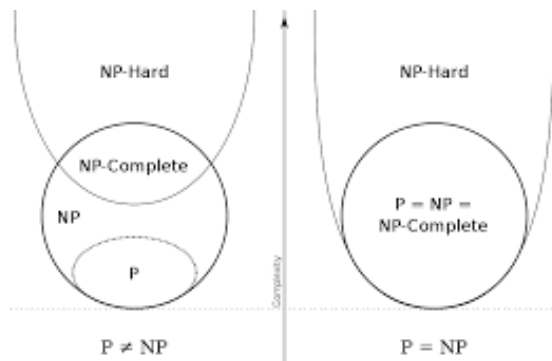
Currently, in practice, any problem that takes polynomial time on the non-deterministic TM can only be achieved in exponential time on a deterministic TM or on a conventional computer. A non-deterministic TM cannot be converted to a deterministic TM. However, non-deterministic finite automata can be transformed into deterministic ones. This has the downside of there being an extremely long run-time and every possible path must be tested. This has the downside of every possible path having to be tested, leading to a long run-time, even if smart processors can exclude a few.

It has been proven that any problem that can be solved by a non-deterministic TM can also be solved by a deterministic TM. However, it is not clear how much time it will take. $P=NP$ means that if a problem takes polynomial time on a non-deterministic TM, then a deterministic TM can be built to solve the same problem, also in polynomial time. It has not been shown that it can be done, but no one has been able to prove that it cannot be done either.

If a problem is in polynomial time, it is in non-deterministic polynomial time. Given a potential answer for a problem in P, the answer can be verified by simply recalculating the answer. However, is every problem in NP in P? If a problem can be verified in polynomial time, can the answer be computed/no solution be found in polynomial time?

P-complete problems are complex and refer to a NP problem X, such that any NP problem Y can be reduced to X by a polynomial reduction. If any NP-complete problem has a polynomial time solution, then every problem whose solution can be verified in polynomial time can be solved in polynomial time and therefore $P=NP$. If a problem can be solved efficiently, anything can be efficiently solved in NP. Therefore if anyone could to prove that

$P \neq NP$, it would be certain that there is no way to solve an NP problem in polynomial time on a conventional computer.



An efficient algorithm for an NP-complete problem would mean that the present assumption of the classes P, NP and NP-complete was completely incorrect, because it would mean that every NP problem (including all NP-complete problems) was actually a P problem.

If $P=NP$, what would it mean?

$P=NP$ could lead to disastrous consequences, as for every efficiently verified solution, the solution could be found just as quickly.^[1] Cryptography would become impossible and consequently any sense of privacy or verifiability on the Internet. The encrypted text and the key could be efficiently used to produce the original text and if $P=NP$, the key could efficiently be found without being known beforehand. Password cracking would become trivial.

These problems could be solved by having non-NP problems - the solutions of these problems cannot be verified using polynomial time solutions and hence the solutions also cannot be found using polynomial time solutions. This would lead to very complex solutions and could therefore be said to be inefficient and take up an unacceptable amount of storage and time.

However, there would also obviously be a huge number of advantages if $P=NP$, but there is much debate on whether the disadvantages outweigh these. Michael Sipser, the head of the MIT Department of Mathematics and a member of the Computer Science and

Artificial Intelligence Lab's Theory of Computation Group (TOC) stated: "When you're faced with a new computational problem, what the theory of NP-completeness offers you is, instead of spending all of your time looking for a fast algorithm, you can spend half your time looking for a fast algorithm and the other half of your time looking for a proof of NP-completeness."^[1] For example, problems such as planning, or resource allocation could be solved effectively and efficiently.

It is generally assumed that $P \neq NP$, that there is no general way to find solutions, alluding to the idea that the time in which a program is verified and that at which it is solved is not equal. However, no proof has yet been established for either $P = NP$ or $P \neq NP$. After decades of studying more than 3000 NP problems, no one has yet been able to find a polynomial-time algorithm for these problems.

A $P=?NP$ poll of sample size 100 in 2002 found that 61 voted that P was 'probably' not equal to NP, whilst only 9 voted $P=NP$. Although these 9 people were a minority, they were all well-respected members of the community. However, a large proportion of these 9 voters confessed that they did this simply to play devil's advocate.^[2]

If you think you can find an efficient general solution formula for any NP-complete problem or disprove that it exists, you could win US\$1,000,000!

Is $P=NP$?

References:

- 1) Hardesty, L. (2009). *Explained: P vs. NP* [online] Available at: <http://news.mit.edu/2009/explainer-pnp> [Accessed 10/07/2020]
- 2) Gasarch, W. I. (2002). *SIGACT News Complexity Theory Column 36. SIGACT News*, 25. [online] Available at: <http://www.cs.umd.edu/~gasarch/papers/poll.pdf> [Accessed 10/07/2020]

The Math Behind Neural Networks

Advait Menon

A neural network is a type of machine learning that is modelled around the human brain. The process by which the brain learns is sometimes remembered with the rhyme “neurons that fire together, wire together”. This refers to the idea that when two neurons fire electric signals simultaneously, the connection between them is enhanced which in turn create memories (a process known as long term potentiation) and are responsible for us learning. Similar to how the brain is made up of approximately 100 billion interconnected neurons, a neural network is a collection of artificial neurons working together in layers to communicate information (although on a much smaller scale than the brain). The three types of layer that make up a neural network are the input layer, the hidden layers (the layers where weights are adjusted) and the output layer.

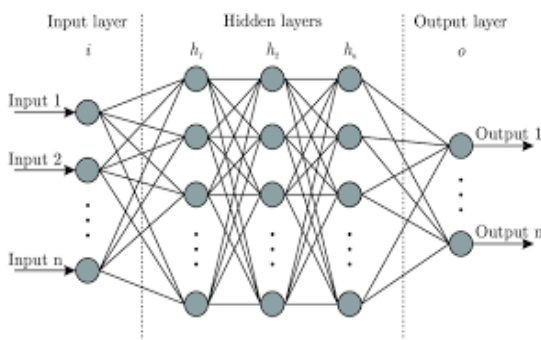


Figure 1. A Neural Network

The smallest and arguably the most important “component” of a neural net is the artificial neuron. The purpose of each individual neuron is to receive a vector, x , of values (the input) and then compute a prediction as an output (known as $\hat{y}$). Each neuron contains a summation function and an activation function. A simple neuron with an input and output layer is called a perceptron. The structure of a perceptron is as shown below:

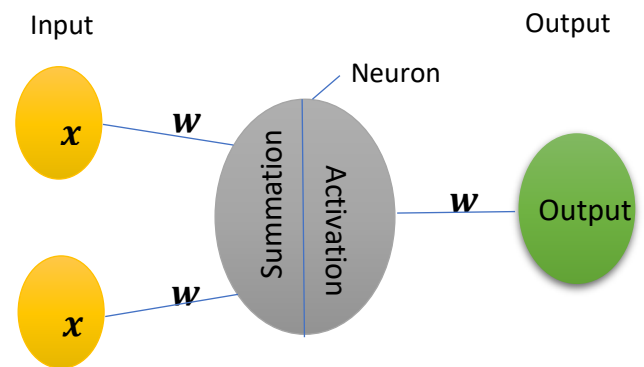


Figure 2. A

The input to the perceptron is in the form of the elements of a vector x . x is made up of “features”. For example, if I utilised a neural net to classify which species an Iris flower belongs to, my x might look like:

$\begin{pmatrix} 1.4 \\ 0.2 \end{pmatrix}$ where 1.4 (x_1) represents the petal length and 0.2 (x_2) represents the petal width.

The features in this scenario are the petal length and petal width. Most perceptrons are not as simple as this one, as well as having multiple layers, they will contain many more features (using the Iris flower example again, other features like sepal length and sepal width would be required to improve accuracy). Each input $x_1, x_2, \dots, x_n$ is assigned an associated “weight”, $w_1, w_2, \dots, w_n$. Weights can be likened to the connections between neurons in our brains. It affects how much influence a change in the input will have on the output. A low weight value will alter the output very little, whereas a larger weight value will more significantly alter the output. A column vector w is initialized before the neural net begins training, consisting of the weights - which are randomized small numbers (that are adjusted throughout the training process).

For a neural net to predict an accurate $\hat{y}$, it must first be trained with a test sample. A test sample consists of inputs (x values) that are representative of a population and the corresponding observed y values. The network will then go through a set amount of iterations, with each iteration consisting of two steps: forwards and backwards propagation.

In forward propagation, the inputs are passed through the network to produce the output. After each input is associated with the corresponding weight, the inputs are fed into the neuron which first applies the summation function. This calculates the sum of the products of weights and inputs and then adds a bias:

$$S = w_1x_1 + w_2x_2 \dots + w_nx_n + b$$

Which can be summarized in sigma notation as:

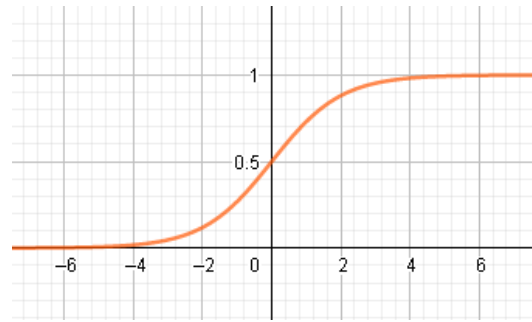
$$S = b + \sum_{r=1}^n w_r x_r$$

Like w , the bias (b) is a randomly initialized before training and is adjusted throughout the learning process.

The output of the summation is then applied over the activation function. The activation function is responsible for converting the weighted sum into the output. Without an activation function, a neural network would just be a combination of linear functions, so it would be unable to learn to approximate more complex functions. Although there are many different activation functions and they can all be used for different applications, the most common are the sigmoid, hyperbolic tangent (tanh) and Rectified Linear Unit (ReLU) functions.

The sigmoid function is defined by:

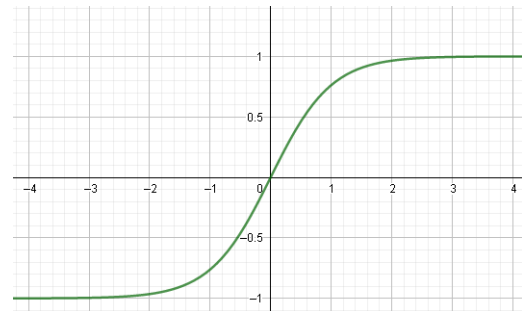
$$f(x) = \frac{1}{1 + e^{-x}}$$



All inputs are mapped to values between 0 and 1. This makes this function especially useful in the output layer if the neural net is being used for binary classification (e.g. checking if something is a dog or not a dog), or if a probabilistic output is needed.

The tanh function is defined by:

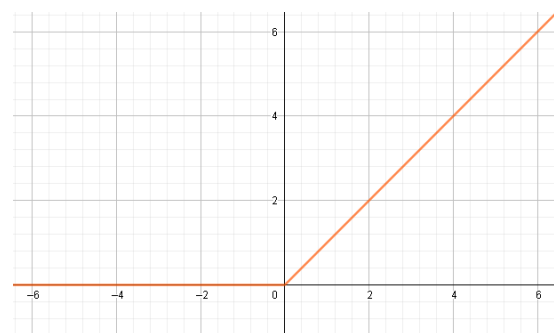
$$\tanh(x) = \frac{2}{1 + e^{-2x}} - 1$$



All inputs from this function are mapped in the range -1 to 1 and centered around zero. It is usually used in the hidden layers.

The ReLU function is defined by:

$$f(x) = \begin{cases} 0 & \text{for } x < 0 \\ x & \text{for } x \geq 0 \end{cases}$$



It maps all inputs between 0 and the greatest value of x . One of its biggest advantages over the tanh and sigmoid functions, is that it does not use up much power to compute and can thus mean reduced training times. It is the default activation function used in the hidden layers.

The output from the activation function is multiplied with a weight (represented by w_3 in Figure 2.) and then supplied to the output layer (which may have its own activation function), to give the predicted $\hat{y}$ value. This occurs in each neuron. In a multi-layered neural network where there are several hidden layers, the outputs are passed on to the subsequent layers where they are again processed by more neurons before finally reaching the output layer.

After predicting $\hat{y}$, backwards propagation starts. In order to “learn”, neural networks need to be able to identify how close their predictions are to the observed y value, this is where the loss function is made use of. The loss function (sometimes called cost function) quantifies the error between $\hat{y}$ and y . The purpose of training a neural network is to minimise the output of the loss function so that by the end, the $\hat{y}$ predictions converge on the observed y . The most commonly used loss function is Mean Squared Error (MSE). MSE is the average of the squared difference between $\hat{y}$ and y and is calculated as follows:

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Where n represents the number of samples in the training data and i is the iteration.

To minimise the output of the loss function, the values of each weight and bias need to be adjusted after each iteration. This is done by calculating the partial derivatives of the loss function in respect to each weight and bias using the chain rule. For example, in Figure 2., the partial derivatives would be calculated as follows:

$$\begin{aligned} \frac{dE}{dw_1} &= \frac{dE}{dA} \times \frac{dA}{dS} \times \frac{dS}{dw_1} \\ \frac{dE}{dw_2} &= \frac{dE}{dA} \times \frac{dA}{dS} \times \frac{dS}{dw_2} \\ \frac{dE}{db} &= \frac{dE}{dA} \times \frac{dA}{dS} \times \frac{dS}{db} \end{aligned}$$

Where E is the loss function, A is the activation function and S is the summation function.

Finally, all the weights and the bias are then updated in each layer according to the following formulae:

$$\begin{aligned} w_{\text{new}} &= w_{\text{old}} - \alpha \frac{dE}{dw_{\text{old}}} \\ b_{\text{new}} &= b_{\text{old}} - \alpha \frac{dE}{db_{\text{old}}} \end{aligned}$$

α represents a hyperparameter known as the learning rate. It is assigned as a small number (normally in the range 0 to 1) and controls how quickly the neural net learns by ensuring the weights and bias do not change too drastically following each iteration.

Once all the iterations have completed and the neural net is trained, it can be used to predict an accurate $\hat{y}$ from inputs it has never seen before during training. This property of neural networks makes them extremely powerful tools and although the math behind them is not too complex, they are utilised in a wide range of applications. Multi-layer perceptrons are used to approximate the functions between variables, even if the relationship is unknown, making them useful in regression analysis. Convolutional and recurrent neural networks are used for image and speech recognition, respectively. Neural networks can also be pitted against each other in a General Adversarial Network (GAN) which can generate realistic images. In the future, neural networks could be used in applications such as self-driving cars, improving stock prediction, music composition, self-diagnosis of medical problems and much more!

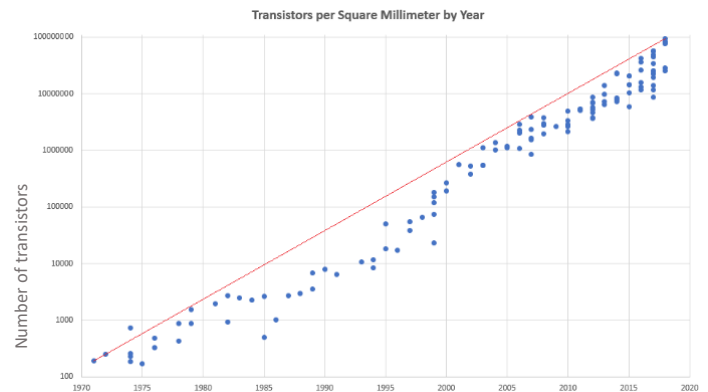
Are Quantum Computers an Opportunity or Threat?

Rishi Tandon

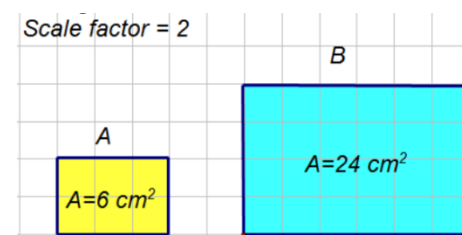
Quantum Computing is a ground-breaking field of technology. Clock speed is a measure of how many instructions can be executed per second. If the clock speed is greater, the CPU can execute more instructions per second meaning that the computer would run more smoothly. By 2025, it is highly possible that we could make quantum computers which can potentially have clock speeds of around a billion times more than today's regular PC's which have clock speeds of around 4.0 GHz (4,000,000,000 instructions executed per second). This is due to research which shows that we could use quantum properties of particles to our advantage. It may seem that there are no downsides to quantum computers, but it also has disadvantages. So, are they a positive or a negative?

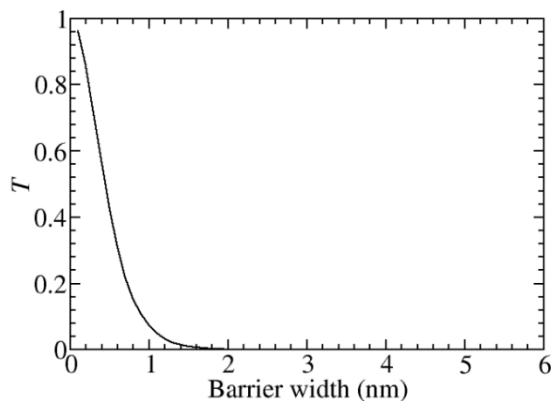
Transistors are responsible for computers working. They are physical parts made from semiconductors which control the flow of electrons – electricity. They are responsible for allowing computers to work through logic gates. By combining logic gates, you can carry out arithmetic operations which is what computers are based off.

Moore's Law currently states that the transistors per square inch should double every two years. This is shown by the image below which shows the number of transistors on chips per square millimetre. The red line displays the predicted number of transistors and the blue points are average data points. The reason the growth appears linear is because the y axis is logarithmic. So far, this trend has been proven to be true.



In order for the trend to continue, the obvious thought would be to continue making transistors smaller. However, in the foreseeable future, these trends may cease to exist due to the size of transistors as they are currently the sizes of atoms. This would allow quantum effects to take part in the transistors, which could lead to degradation of the transistors. The main quantum effect that currently takes place is quantum tunnelling. Quantum tunnelling is a problem because transistors are like light bulbs, as they have an on or off value, which equates to a value of 1 or 0 in the case of binary. Quantum tunnelling is the phenomenon through which electrons can phase through a barrier if it is thin enough. This means that electrons could flow, causing an electric current which would register as a value of 1 on a hard disk, leading to transistors being ineffective as they depend on logic gates, in order to carry out operations. The graph below shows the probability of quantum tunnelling occurring. For reference, current transistors are 14nm so it is very unlikely that current computers would have quantum computing be a problem. The scale factor would have to be smaller by a factor of $\sqrt{2}$ as the scale factor would become smaller due to the way that area changes (shown in the figure below).





By doing the maths, you can work out the number of years by which there is a reasonable chance of 0.1, which may seem small but is quite large due to the vast number of electrons passing through the transistor per second.

$$14 / (\sqrt{2})^n \leq 1$$

$$14 \leq (\sqrt{2})^n$$

$$n \geq 2 \log_2 (14)$$

$$n \geq 7.6$$

$$n = 8$$

This means that after 16 years computers would become unviable if Moore's Law still holds. Quantum computers could potentially solve this problem. Conventional computers work by using bits, sequences of 1's and 0's depending on transistors whereas. Quantum computers would work by using qubits which could involve quantum properties such as particles of light or the spin of electrons. Light has a varying polarity, which is the direction of the vibrations that are occurring. By putting it through a polarization filter, it 'decides' whether it is horizontally or vertically polarized. These 2 states could equate to a 1 or a 0. The spin of an electron is the angular momentum of the electron. Depending on whether it spins clockwise or spin up, or anticlockwise, which equates to spin down. The process of measurement is similar to that of the polarization of light.

Quantum computers are much faster than conventional computers because quantum particles can hold values of 1 or 0

simultaneously as they only hold a value when the value is measured. This phenomenon is known as superposition which is the idea that when the state is unmeasured, the state of the particle is in both states simultaneously. This means that when not being measured, 4 qubits can hold 16 values simultaneously whereas conventional computers require 4 bits, of which only 1 of the values will be used.

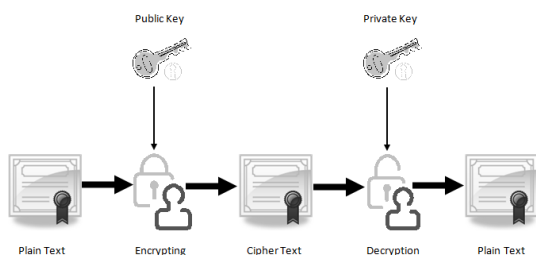
There is still an opportunity as we could use quantum computers to create a new type of security which would be more secure than our current network security. They could also be used in advanced modelling situations such as for quantum physics or improving nanotechnology due to the immense processing power required. This could lead to numerous lives being saved due to the discovery of new drugs. Scientists would be able to model the interactions of new drugs with cells to find cures to diseases, which would save lives. Quantum computers have huge potential for the future of nanotechnology.

Humans have currently made a working quantum computer in IBM which has 50 qubits. It is currently used as a cloud computing service, due to its immense computing power. This system is currently used in developing countries such as Senegal. It is helping 16 African Universities to encourage thought and ideas without cost being an issue for developing countries. This is being done in order to allow thought and innovation without wealth being a factor if countries don't have the resources to carry out certain required calculations, which would also help to bridge the development gap.

The quantum computer would also have to be super cooled, due to the high temperatures involved. This would most likely be done with liquid nitrogen which would be about \$36,500 for cooling, assuming the computer is in use 24/7 for a year. The reason the temperatures are so high is due to the high clock speed. If the clock speed is greater, you would require

cooling as the temperature would otherwise get too great. The electricity bill would also be expensive, around \$25,000 for a year. This money could be spent elsewhere in finding ways to reduce the effect of quantum tunnelling.

It would also result in the destruction of network security as the keys are currently based on multiplying 2 very large prime numbers, as the prime decomposition of the larger number will only have 2 factors. Currently, it takes modern day computers thousands of years to be able to find these 2 prime numbers. However, quantum computers will greatly reduce the time taken to find these 2 numbers, reducing internet security. The private key code would not be able to be cracked as it is individual to each person so the data should still remain safe. The diagram below displays an example of how modern-day encryption works. Currently, everyone can see your public key, in the form of a product of 2 large prime numbers, but only you have access due to your private key. This would mean your data would be safe as the complete decryption process would not be able to occur. In order for quantum computers to allow network security to work safely, a new method of encryption would have to be made which could use quantum computers to their advantage.



In conclusion, a quantum computer would be a benefit to humanity as long as new encryption methods are developed. However, as technology improves, these problems should be solved. The cost is also a major factor, but this is worth it for the much greater processing power. It would lead to

increased life expectancy if it was used to test drugs and would mainly be more useful in nanotechnology and other tasks which require high processing power. Quantum computers would be useful by the end of the 22nd century as humans could work on space travel in order to get closer to becoming a type 3 civilisation. Therefore, I believe quantum computers are an advantage.

Encryption: The Quantum Mechanics of Communication

Lucy Stock

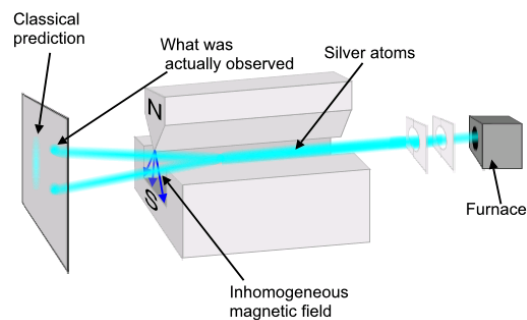
Quantum physics fundamentally shapes our world, through the way that quantum mechanics influences how the universe works and how the constituent chemicals of our world are formed and interact. As massive leaps in our study of quantum mechanics are made, this progress can also be utilised to better our world. One area in which developments in quantum physics has had a great impact is communication.

Physical phenomena

The quantum binary integer, or qubit, is used in quantum computation and allows quantum computers to be potentially staggeringly more advanced than classical computers. The two particles used to represent qubits, in different implementations of quantum computing, are electrons and photons. They are unique, in that the spin of an electron and the polarisation of a photon are either one thing or another when measured; an electron has “up” or “down” spin and a photon can either pass or not pass through a polarised filter. These are the 1s and 0s required for computation - the binary aspect. However, their uniqueness is that before they are measured, they are in what is called a superposition of these states. They are not 1 or 0, they are a combination of 1 and 0 weighted by the probability that they will become 1 or 0 when measured. [1]

So how does the spin of an electron work as a bit? The Stern-Gerlach experiment demonstrates this. Silver atoms have 47 electrons and all but the one in the outer orbit are paired. The pairs spin in opposite directions and so the magnetic field caused by the movement cancel out, leaving the field

created by the lone electron which turns the whole atom into a magnet with a north and south pole.



[3] The Stern-Gerlach apparatus.

The Stern-Gerlach experiment set out to determine whether this axis could have any direction or if it was constrained in some way. They sent a stream of silver atoms through a pair of magnets toward a screen. The magnets were specially shaped into a V shape to make the south magnet act more strongly than the north and so if the pole of the atom was vertically aligned, the south magnet would act more strongly causing the particle to be deflected. It was expected that a line would form on the screen as all the atoms' axes were in different directions and so were attracted/repelled by different amounts. In actuality, only two dots formed - one at the extreme top and the other at the extreme bottom. This meant that every single atom was vertically aligned - none had another orientation. This observation demonstrated the spin of the electron was quantised. Electrons also act like magnets and when you determine the spin (the standard terminology for determining the axis of the magnetic poles) they are either “spinning” one way or the opposite way. However, we cannot repeat this experiment using just electrons themselves, as they are a charged particle and so the beam of moving electrons would be repelled by the magnets. Whenever we measure our electron, whatever angle the magnets are rotated to, they will either go in the direction of the south magnet or the north magnet - yes or no, 0 or 1. Thus, electron spin

is quantised and so can be used as a binary value. [1]

Another important aspect of measuring spin occurs when it is measured in multiple directions. If you have a detector measuring in the vertical direction and have subsequent detectors measuring in the same direction, the half of the electrons deflected up by the first will also be deflected up by the second, and the other half that are deflected down will be deflected down by the second as well. If instead the second set of detectors are measuring at 90° to the first detector, exactly half of the “up” electrons will be deflected left and the other half will be deflected right, in a completely random order and there is no way to predict which way the electrons passing through would deflect. If there was also a third set of detectors measuring vertically again, exactly half would be deflected up and the other would be deflected down. This is regardless of their initially measured spin. Some electrons that we measured with vertical spin N (their north pole is pointing upwards) will now have vertical spin S. This shows that measurements affect outcomes. When you measure the spin of an electron it changes from a superposition of possible states to definitely one of those states and so the value of the electron is irreversibly changed. [1]

The other particle used to represent qubits, the photon, has a much more easily visualised binary result. Their yes or no, 0 or 1 value is whether or not the photon can pass through a polarised filter. Photon polarisation is also quantised. Photons are polarised in two perpendicular directions, which are both perpendicular to the direction of travel. If the photon is polarised in the same direction as the filter, it will pass through. If not, it is at 90° to the filter and it will not pass through. Sending a photon through the filter is measuring its polarisation - by shining some light through a pair of polarised sunglasses, you are performing a quantum measurement. If we have two filters polarised in the same

direction, overlapping them will have very little effect on the intensity of the light coming through. However, if the second filter is at 90° to the first, all the photons that are emitted from the first filter are confirmed to be perpendicular to the second and so no light passes through. If we add a third filter between the other two at 45°, we know the photons passing through the first filter are vertically polarised and then the second filter is now measuring if these photons are polarised at 45° or 135°. Half of the 0° photons will now be polarised at 45° and half will be at 135° and the ones polarised at 45° will pass through. At the final filter, which is measuring if they are polarised at 0° or 90°, half of these 45° photons will jump to 0° and the other half will become 90° and will pass through. This means 1/8 of the light that hit the first filter makes it out the other side. More light passes through 3 filters than 2. [1]

Mathematical explanation

The mathematical models I am using to describe the spin of an electron are 2 dimensional rather than in 3 dimensions so the models may differ from those from other sources. The two potential directions for the spin of an electron or the polarisation of a photon are represented with an ordered orthonormal basis of vectors. These are perpendicular unit vectors and there are as many vectors as there are dimensions we are working in. The standard 2-dimensional basis is $\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$. An ordered basis is when the order of the two vectors in the basis matters and this is true when we are considering qubits, as the first vector is considered a 0 and the second is 1. When bases are ordered they are notated with circular brackets e.g. $\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right)$ and this is the notation I will use throughout. I will also use bra and ket notation to represent the vectors. We can write every vector as a combination of the basis vectors we are using i.e. $\begin{bmatrix} x \\ y \end{bmatrix} = n \begin{bmatrix} a \\ b \end{bmatrix} + m \begin{bmatrix} c \\ d \end{bmatrix}$. This is how we work out which vector an electron will “jump” to when

measured in a different basis. n and m are what is called the probability amplitudes, the square of these numbers gives the probability of the electron jumping to that vector when measured in that basis. [1]

To form the basis vectors for any direction of measurement for electron spin, just substitute the angle a from the vertical we are rotating the measurement apparatus by, into $\begin{pmatrix} \cos(\frac{a}{2}) \\ -\sin(\frac{a}{2}) \end{pmatrix}$, $\begin{pmatrix} \sin(\frac{a}{2}) \\ \cos(\frac{a}{2}) \end{pmatrix}$. [1]

For photon polarisation, angle a is the angle past the vertical of the rotation of the filter and the basis is $\begin{pmatrix} \cos(a) \\ -\sin(a) \end{pmatrix}$, $\begin{pmatrix} \sin(a) \\ \cos(a) \end{pmatrix}$. [1]

A qubit has the form $d0|b0\rangle + d1|b1\rangle$ where $d0$ and $d1$ are the probability amplitudes of the qubits “jumping” to 0 and 1 respectively. Although probability cannot be negative, probability amplitudes can, and this allows probability amplitudes to work much in a similar way to physical waves. They can interfere and superpose. If the waves are in phase and their peaks and troughs line up, they constructively interfere resulting in a greater amplitude. If they are out of phase, the waves destructively interfere, decreasing the amplitude of the resulting wave and even completely cancelling it out in some cases. We can take advantage of this in quantum algorithms as we can choose linear combinations where the result interferes in such a way that the incorrect or useless terms destructively interfere and cancel, and the terms desired are amplified. For example, if we have the two “horizontal” qubits $\begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$ and $\begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix}$, and measure them in the standard basis, they will jump to either $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ or $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ with equal probability - translating to either 0 or 1 with equal probability. However, if you take a superposition of the two original qubits $|v\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} + \frac{1}{\sqrt{2}} \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix}$ the probability when measured in the horizontal direction is 50/50 (note how $(\frac{1}{\sqrt{2}})^2 = \frac{1}{2}$ showing this) and so

would get either 0 or 1 with equal chance.

Whereas when measured in the vertical

direction we are certain to get 0 as $|v\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} + \frac{1}{\sqrt{2}} \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix} = 1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + 0 \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. The terms that give $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$, which is the first vector in our vertical ordered basis (our binary 0), have interfered constructively and the terms that give 1 have interfered destructively. An advantage of quantum computing is we can construct our equations to take superpositions. This enables it to calculate many things simultaneously, and then we can set it up so incorrect answers destructively interfere and the correct, relevant answer, is amplified with constructive interference. [1]

Computational application

The fact that qubits “jump” to either 1 or 0 when measured is incredibly useful. It can be used to check whether or not communication is being intercepted. If person A with orthonormal basis ($|a0\rangle, |a1\rangle$) wants to send data to person B ($|b0\rangle, |b1\rangle$), they can use a stream of qubits to send this information. Person A uses their measuring apparatus to sort qubits into $|a0\rangle$ and $|a1\rangle$ and sends the one relating to the bit of data desired. However due to their different bases, there is no guarantee that when person B measures, they will get the same result as person A. To find what happens, we calculate the probability amplitudes of the state to which the qubit will jump when person B measures it. $|a0\rangle = d0|b0\rangle + d1|b1\rangle$ meaning for any 0 sent by person A, person B will measure 0 with a $d0^2$ chance and 1 with a $d1^2$. [1]

Take the example of the two measuring at 90° to one another: person A's basis being $\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

and person B's basis being $\begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}, \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix}$. $\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} + \frac{1}{\sqrt{2}} \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix}$ so for any 0 sent by person A, B would receive 0 or 1 with a 50/50 chance - half the time they would get it wrong without knowing which ones are wrong. [1]

If person A measures in the 240° direction, using basis $(\begin{bmatrix} \frac{1}{2} \\ -\frac{\sqrt{3}}{2} \end{bmatrix}, \begin{bmatrix} \frac{\sqrt{3}}{2} \\ \frac{1}{2} \end{bmatrix})$ which can be simplified to $(\begin{bmatrix} \frac{1}{2} \\ -\frac{\sqrt{3}}{2} \end{bmatrix}, \begin{bmatrix} \frac{\sqrt{3}}{2} \\ \frac{1}{2} \end{bmatrix})$ since multiplying a ket by -1 is an equivalent ket and person B measures in the 120° direction, using basis $(\begin{bmatrix} \frac{1}{2} \\ -\frac{\sqrt{3}}{2} \end{bmatrix}, \begin{bmatrix} \frac{\sqrt{3}}{2} \\ \frac{1}{2} \end{bmatrix})$, we can also calculate the probability to which person B gets the correct information. If person A sends 0, they send the qubit $\begin{bmatrix} \frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{bmatrix}$. To find the probability amplitudes we form a matrix from the bras of the basis vectors of person B and multiply the qubit by this. $\begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} \frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{bmatrix}$. This means that $\begin{bmatrix} \frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{bmatrix} = -\frac{1}{2} \begin{bmatrix} \frac{1}{2} \\ -\frac{\sqrt{3}}{2} \end{bmatrix} + \frac{\sqrt{3}}{2} \begin{bmatrix} \frac{\sqrt{3}}{2} \\ \frac{1}{2} \end{bmatrix}$. When person B measures the bit sent as 0, they will get 0 with a probability of $1/4$ and 1 with a probability of $3/4$. This is the same for all cases where they choose from 3 bases where the third is the standard basis and end up with different bases - person B receives the correct bit with a probability $1/4$. [1]

There is a reason that the two people do not just agree on a single basis. With the same basis, if person A sent a 0, person B would receive 0 with certainty, and same for sending 1. However, since they are trying to ensure their data is secure, if someone intercepting with malicious intent, person C, was also using the same basis, they would all receive the same data, person C would know everything that person B was learning. This is of course the opposite of what we want to happen. [1]

To keep the communication secure, we want to encrypt it. This works by the two communicating parties deciding on a key, a long string of bits, and then using this key to encode and decode the messages. The key is required to decode a message - decryption is carried out by working out the key and then decoding the message. People trying to intercept the communications need the key - so rather than trying to work out the key, it is often easier to intercept the original key communication. This is why it is important this

original message is incredibly secure. The BB84 protocol is an encryption method that uses qubits to let the communicators know if their key is being intercepted. It uses two sets of ordered orthonormal bases: the standard $(\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix})$ which we will denote as V as it is spin in the vertical direction, and $(\begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}, \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix})$ which we'll denote as H. As usual, 0 is the first vector and 1 is the second. Person A chooses the key, a string of classical bits, length $4n$. For each bit of the key, they choose one of the bases, V or H, with equal probability at random. They then send the qubit corresponding to both the bit and the basis. For example, 0V would be $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and 1H would be $\begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$. They record the basis used for each bit. Person B, when receiving the stream of qubits, will choose the basis to measure in with an equal probability at random. They will then record two strings, the string of bits and the string of the bases they were measured in, both length $4n$. If they have both chosen the same basis, person B will receive the bit sent by person A with certainty. If they chose different bases, person B will receive the correct bit only half the time - so no information is transferred. They compare their strings of Vs and Hs over an unencrypted line and keep the bits where they chose the same basis and discard the bits when they chose differently. Since they both chose randomly, the remaining string will be $2n$ long. Now they must check if their transmission has been intercepted. If they had been intercepted, when person C measures the qubit sent by person A it would change the qubit into the basis vector of the basis they choose. Half the time person C will also agree with A and B meaning they all get the same bit, but the other half of the time person C will disagree, changing A's qubit into a superposition of their basis states meaning when B receives the bit there is an equal chance of getting a 0 or a 1. This means that if the message is being intercepted, $1/4$ of the time person A and B will get the same bit, but $1/4$ of the time B will receive the wrong bit due

to C's measurement. A and B compare $\frac{1}{2}$ of their remaining bit string over an unencrypted line, sending a string length n and keeping a string length n secret. If they are being intercepted, $\frac{1}{4}$ of the bits they sent will not match up - they know someone is listening in and they should find another way to communicate. If their communication is private, they will agree on all bits sent and they can use the remaining n private bits as the key. This means that they can be secure in the knowledge that the only way that people would be able to break their encryption is to work out the key - an impossible task. [1]

Well, almost! Although developments in quantum computation can increase security, it can also be a major security threat. Most secure data transferred over the internet currently is encrypted with a massive number, called an RSA number, and it is decoded using the two prime numbers that multiply to make this number. For people intercepting this message who do not have the two primes that multiply to make N , it would take years of processing time on a classical computer to find these numbers, which is completely unfeasible. All they can do is guess every single prime number until they find a factor of N . RSA 768 took 2000 years of processor time to crack. Shor's algorithm is a quantum algorithm that takes advantage of quantum superposition and interference to find these factors quickly. It guesses a number and then refines this guess - trying to find a guess g where $N=gh$ or $N=ab$ and $g=ac$ (in this case we can use Euclid's algorithm to find the common factors and thus find a). Once you have one factor, you just divide N by this factor and the encryption is broken. [2]

The complex quantum part is refining guess g into a better guess. If you have numbers A and B which share no common factors, if you multiply A by itself enough times, the result will be one more than the product of B and something else. $A^p = nB + 1$. This means for any guess g and any number N that do not share factors, $g^p = mN + 1 \Rightarrow g^p - 1 = mN \Rightarrow (g^{p/2} + 1)(g^{p/2} -$

$1) = mN$. The refined guess is $g^{p/2} \pm 1$. This will either be a factor, or due to the m , a multiple of the factor. However, there are some problems with this $g^{p/2} \pm 1$ could be a multiple of N and so would not be useful for finding prime factors. If p is odd, $g^{p/2}$ will not be a whole number and we are looking for integers and so would not be correct. For a random guess g , 62.5% of the time either one or both of these situations occur, so the guess is useless. In the remaining 37.5% we can use our random guess g to give a factor of N and break the encryption. This means with fewer than 10 guesses we are 99% certain to break this encryption. However, the process of converting g into $g^{p/2} \pm 1$ takes a long time on a classical computer so we cannot use this method currently - it is actually faster to just guess every single prime until we find the number we are looking for. Where a regular computer can only give one result for one calculation at a time, a quantum computer can calculate many things at once by performing the calculation on a superposition of values and would output a random one of these results, each having a different probability. On top of this, if the result that is output could have come from more than one value, you are left with a superposition of all the values that could have given this output - making it even easier. The reason quantum computers are so fast is that instead of outputting a random value, the calculation can be set up so even if you are calculating every single possible answer, incorrect answers destructively interfere and the correct answer is the only one that can possibly be output. When trying to find power p , we guess many powers x and raise g to this power, and the quantum computer will output $|x, g^x\rangle$ for all x . Then it calculates how much bigger than a multiple of N g^x is and outputs the remainder $|x, r\rangle$. When we send in a superposition of numbers, this calculation is performed on all simultaneously and a superposition of powers g could be raised to and the remainders r is outputted. If this is measured, however, we will get a random

output $|x, r\rangle$ with no guarantee $r=1$ so we must make all incorrect guesses destructively interfere so the only possible output is $|p, 1\rangle$. We use the fact that $g^x = mN + r$ and $g^{x+p} = yN + r$, any numbers whose powers differ by p will all have the same r - p has a repeating property. If we just measure $|r\rangle$, we are left with a superposition of all powers x that give r , and these x values will all be p apart. But how do we find p if we cannot measure more than one? Well, this means they repeat with a frequency $1/p$ and we can use a Fourier transform (a function that takes a wave and transforms it into a graph of the frequencies that make it up) to find the frequency $1/p$ and just invert it to get p . The quantum Fourier transform works such that for any number inputted as a suggested frequency, it will output a superposition of every number, weighted where the weights form a sine wave with the suggested frequency. The higher the inputted number, the higher the frequency of the produced sine wave. Since we input a superposition of numbers, the output is a superposition of superpositions, the sine waves will interfere and add together or cancel out. If our input superposition has all number separated by amount p , the sine waves cancel out such that the final sine wave has frequency $1/p$, solving our problem. We can then invert p , and if it is even, we can raise $g^{p/2} \pm 1$ and as long as this isn't a multiple of N , our refined guess will share factors with N . Euclid's algorithm can then be used to find these factors and we have broken the encryption. [2]

However, this is a way off from being code-breaking reality. Current quantum computers can only hold a few bits, meaning we can currently only factorise numbers up to about 35 with this algorithm - nowhere near those used in encryption. There are other methods, like adiabatic quantum computation, that have factorised numbers as big as 291311 with only 3 bits, but even the smaller numbers used in encryption like RSA 230 would need 5900 bits even using this vastly more complex

algorithm. Shor's algorithm works in theory and may be a threat in the future, but we are a long way off having the hardware to support it being any threat to data security. [2]

This is only a brief glimpse into one aspect of how quantum computing is developing. The theory and the processes behind incredible technology is just ready and waiting for new generations of engineers and computer scientists to develop the hardware we need. Physics is carving the way for extraordinary, impactful and even sometimes slightly scary developments that will change how we communicate in the future.

References:

- [1] C. Berhardt, Quantum Computing for Everyone, The MIT Press, 2019.
- [2] M. Physics, "How Quantum Computers Break Encryption | Shor's Algorithm Explained," 30 April 2019. [Online]. Available: <https://www.youtube.com/watch?v=lvTqbM5Dq4Q&feature=youtu.be>. [Accessed 22 May 2020].
- [3] J. D. Best, "Alternative Explanation of the Stern Gerlach Experiment," 2015. [Online Image]. Available: http://vidainstitute.org/?page_id=489. [Accessed 4 June 2020].

Allied Cryptosystems in the Second World War

Oliver Wright-Jones

School of Engineering, University of Warwick

<https://youtu.be/yJz4ctVKual>

As war was raging all across Europe, with both sides trying to outmaneuver the other in territorial games on the continent, it became increasingly important to intercept the enemy's communications. As is famously known, the German Enigma machine was the cryptosystem used by the Axis Powers in communications starting from the mid-1930s. Adapted from the Commercial Enigma A26 [1], the Military Enigma was sufficiently complex as to be unbreakable by Hitler's High Command, an attitude maintained until the end of the war. While the Enigma was successfully broken by mathematicians working for the Polish Cipher Bureau as early as 1939 [2] (and so to correct a common misconception: Alan Turing did not break the Enigma), the addition of two more coding wheels (German: Walzen) meant that additional work needed to be done during the war that built on the work of the Polish. Poor operational procedures from German regimental officers and civilian agencies allowed for the reverse-engineering of the ciphers by the British Government Code and Cypher School at Bletchley Park.

Perhaps less known by history is how the Allied forces were keeping their communications encrypted. The British had their own adaptation of the Enigma, called the Typex, developed after an inter-departmental committee of GC&CS was formed to study communications of other countries. In January 1935 the committee recommended the construction of Enigma-style machines, almost exactly copying the commercial Enigma, along with some modifications [3].

The main difference was the irregular motor movement (compared to the Enigma's predictable and infrequent movements), with each rotor containing irregularly-spaced indents that would also turn the neighbouring rotor. This completely eliminated patterns in ciphertext, which was the main weakness of the Enigma.

Operators were on average able to encrypt or decrypt 20 words per minute. While this may seem slow, this was offset by the ease of use when compared to the Enigma. Most Enigma machines needed two operators one to input text to the machine, and another to copy down onto paper the enciphered or deciphered characters as the first operator called them out. Typex only required a single operator, and also completely removed the chance of copying errors, as the enciphered or deciphered text was automatically printed on paper tape. Furthermore, while Enigma messages needed to be written, enciphered, transmitted (by Morse), received, deciphered, and written again, Typex messages were typed, enciphered, and transmitted all in one step [4]. It was these inconveniences that caused German operators to start cutting corners and fall into lazy operating habits.

7 separate iterations were developed during the years to 1950, each making minor security and portability improvements, with an estimated 12,000 Typex machines having been built by the end of the war [5].



Figure 1: Operational Typex machine, Government Communications Security Bureau, New Zealand

Of greater interest is what was used by the United States, since this was an even larger deviation from the Enigma design. The ECM Mark II (called SIGABA by the US Army) was the main cryptosystem used through the Second World War and during the early years of the Cold War up to the late 1950s. The letters were not selected by rotors that turned between one letter and the next, rather another bank of rotors, called the control rotors, was used to determine which of the cipher rotors would advance [6]. This meant that just like the Typex machines, any rotor could move at any time, although some moved more often than others. A third bank of five 10-contact rotors, called index rotors, acted as stators that never moved during encipherment. The stators created a complex electrical maze for signals to travel through. The outputs from control rotors were connected to the inputs of the index rotors, and subsequently those outputs were connected together in five pairs, providing the electrical signals that controlled the advancement of the cipher rotors, as in Figure 2.

According to the NSA, over 5,000 machines were deployed by 1943, and well over 10,000 by the end of the war [7]. They were incredibly bulky and fragile, making them near-impossible to use in the field apart from in control rooms in contrast to Enigma, which was far more practical. SIGABA machines were not allowed in foreign territory unless American armed personnel were able to guard them, leaving German codebreakers clueless as to how to proceed with decipherment attempts.

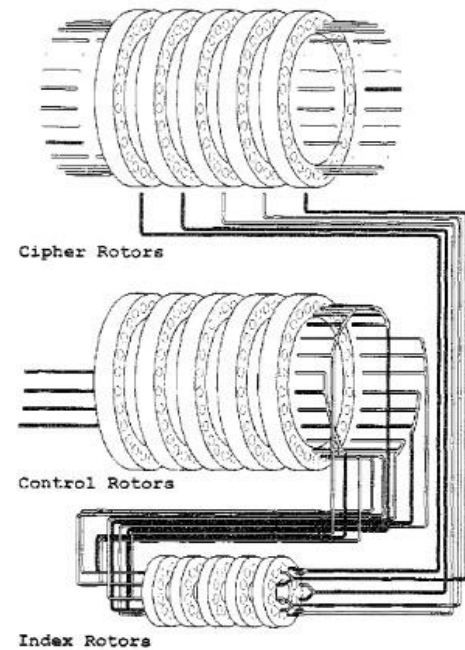


Figure 2: Electrical connections of the rotors in the ECM Mark II

A more intriguing part in the history of both of these machines was how the Allied nations sent messages to each other while using different machines. From 1943, the Americans and the British agreed upon a Combined Cipher Machine (CCM). The British Typex and American ECM Mark II could be adapted to become interoperable by building specific attachments for them. While Typex was shown to the United States, the US Armed Forces never permitted the British to see the ECM, which was a more complex design and thus more valuable [5]. The CCM was initially used on a small scale for naval use from 1 November 1943, becoming operational on all US and UK armed services in April 1944. German codebreakers had no success with the Allied systems despite being able to capture a Typex machine at Dunkirk during the Battle of France, although there were specific cases where the encryption pattern had a period of 338 - and with a long enough message this could have easily been decrypted.

The machines were retired during the 1950s by the Americans and the British as more complex electrical machines were developed, although the New Zealand military were using Typex until around 1973. Both the ECM Mark II and the Typex have remained impervious to cryptanalysis both during their time of operation and in later attempts.

References:

- [1] "Enigma D."
<https://www.cryptomuseum.com/crypto/enigma/d/index.htm>. Accessed: 2019-12-01.
- [2] M. Rejewski, "How Polish Mathematicians Broke the Enigma Cipher," *Annals of the History of Computing*, vol. 3, pp. 213–234, July 1981.
- [3] L. Kruh and C. A. Deavours, "The Typex Cryptograph," *Cryptologia*, vol. 7, no. 2, pp. 145–166, 1983.
- [4] John R. Ferris, *Intelligence and Strategy: Selected Essays*. Routledge, 2007.
- [5] R. Erskine, "The Admiralty and Cipher Machines During the Second World War: Not So Stupid after All," *Journal of Intelligence History*, vol. 2, no. 2, pp. 49–68, 2002.
- [6] J. J. G. Savard and R. S. Pkelney, "The ECM Mark II: Design, History, and Cryptology," *Cryptologia*, vol. 23, no. 3, pp. 211–228, 1999.
- [7] Timothy J. Mucklow, *The SIGABA/ECM II Cipher Machine: A Beautiful Idea*. Center for Cryptologic History, 2015.

To What Extent Has Digital Rights Management Been Effective in Preventing Copyright Abuse?

Oliver Wright-Jones

School of Engineering, University of Warwick

1: Abstract

Copyright has been enshrined in international law since the 1886 Berne Convention [1], and one of the rights of a copyright owner is the right to implement techniques preventing copyright circumvention. A group of techniques popular in the software and entertainment industries is Digital Rights Management (DRM), which typically involves installing accompanying software to protect content from being copied and distributed over a network [2]. It has drawn criticism from consumer groups such as the Electronic Frontier Foundation for being too restrictive and interfering with users' rights under copyright law [3], although certain industry giants claim it is justified after widespread piracy through platforms such as Napster [4]. An example of DRM in the music industry will be analysed, followed by the more general case. Most case studies will be from the UK and US, where the relevant laws are the Copyright and Related Rights Regulations 2003, and the Digital Millennium Copyright Act 1998 respectively. The aim of this essay is to determine whether DRM, and its enabling laws, is effective enough to deter the majority of consumers from committing acts of piracy, or whether a medium needs to be negotiated between companies and consumers as an alternative solution where DRM has failed.

2: Discussion

Companies and individuals have the right to protect their Intellectual Property (IP). In the US, Section 8 of the US Constitution states that "Congress shall have the power to promote the Progress of Science and useful Arts, by securing for limited Times to Authors and Inventors the exclusive Right to their respective Writings and Discoveries" [5], allowing authors and inventors to make profits or exploit their creations ("exclusive Right") while at the same time providing public good ("Progress"). Similarly, in the UK, laws such as the Statute of Anne made copyright "a trade-regulation concept to promote learning and to curtail the monopoly of publishers" [6]. It enabled authors to have more control over the distribution of their works, while also allowed the public to benefit. In this way, it can be said that IP rights and laws are a way to encourage innovation among authors and researchers. At its core, there is a balance of public and private good [7] - the public should have the right to access creative and literary works, but creators should have the chance to make profit or otherwise take advantage of their work. Software and music, the main types of content to be protected by DRM in the Internet age, is intangible property with monetary value. Companies must protect their assets against theft, otherwise they could easily go out of business. Individuals, too, deserve not to have their work stolen or used without permission. Copyright theft prevents authors from the rights to their works and can be a danger profits they are entitled to. The 1886 Berne Convention ensured internationally that copyright theft was a crime against the author and works had the right to be protected. Unfortunately, in the modern age, online piracy of digital media, has always been an issue [7], and this provides an incentive for companies to implement DRM. Although companies could litigate against individuals who use pirated software or stolen music, the nature of the

internet makes it difficult to track specific people, not to mention the time and money it takes to set up a lawsuit. It is therefore easier to implement software solutions to prevent widespread abuses.

Despite this, in the past, direct litigation of individuals has been carried out en masse. In a response to the growth of peer-to-peer (P2P) networks such as LimeWire, Grokster, and Kazaa (to name but a few), the recording industry started taking legal action against the particular users of the networks, by issuing subpoenas to Internet Service Providers for IP addresses they had identified as sharing audio files. By the end of 2007, at least 30,000 people had been sued by the member companies of the Recording Industry Association of America (RIAA) [8]. Most lawsuits sought an amount between \$3,000 (£2,400) and \$11,000 (£8,900). Targets had little choice but to settle, since it would have been more expensive to hire a lawyer. The effect on some individuals was devastating. The campaign was going after a few people at random, seeking a disproportionate punishment, resulting in RIAA being accused of “unnecessary roughness” by the EFF [9]. At its peak, Napster had 80 million users [10], with millions more on other networks that were popular at the time, yet only a few were being sought after by RIAA. Despite the industry’s efforts to deter people from using P2P networks, Internet traffic was not diminished during the litigation period, suggesting that their efforts were, in the most part, futile [11]. However, the wide publicity surrounding the lawsuit campaign managed to increase public awareness of copyright law, even though warnings and threats were mostly ignored by young people, who continued to pirate media [12].

Even though direct litigation proved ineffective and forced the industry to research other methods, there are certain problems with the software-based approach creators and authors, be they companies or individuals, employ to protect their works

from copyright abuse. An important point that groups such as the Open Rights Group make is that anti-circumvention laws, a fairly recent development in UK and US law, distort traditional trade-offs in copyright cases [13] and give a disproportionate amount of power to the industries. Firstly, DRM locks content to the operating system or hardware platform that the author writes their scheme for. This means that consumers cannot transfer content between the devices they own as they might have been able without DRM. Secondly, some schemes such as persistent online authentication require an internet connection to check if a user has purchased the content lawfully, restricting the conditions under which one can enjoy the content they have purchased. Furthermore, copyright laws such as the DMCA contain certain clauses which can be rather dangerous and have unfair consequences for consumers. U.S. Code Title 17, Section 1201, which is globally applicable under the terms of the Berne Convention, states: “No person shall circumvent a technological measure that effectively controls access to a work protected under this title” [14]. While this seems like a reasonable law to make, it prevents one from publishing weaknesses, security or otherwise, in DRM systems. This could mean that software with DRM could be introducing an attack vector for adversaries, and researchers would be prevented from making it public for fear of prosecution. One notable example of the conflict arising from this law was in 2005, an incident known as the “Sony rootkit”. Sony BMG, which owns well-known labels such as RCA Records and Columbia Records, attempted to implement its own copy protection methods on about 22 million CDs [15]. While not a problem in and of itself, the additional software was not part of the EULA when the disk was inserted and ran in the background taking up user resources. Furthermore, there was no uninstaller, and created security holes that could be exploited by malware [16]. Anti-virus firm F-Secure noted in a report on the software that “it is

very inappropriate for commercial software to use these techniques” [17]. While the original researcher, Mark Russinovich, was brave enough to have published his research onto his blog as soon as he discovered the rootkit, it is not a stretch to imagine countless other security vulnerabilities from different software having had delays in their reporting due to researchers being afraid of being stung by DMCA section 1201. This sets a dangerous precedent and unfairly discriminates against consumers, who have a moral right to know whether the software they install on their computer has vulnerabilities.

It is worth examining why certain DRM schemes have been unsuccessful in the past in order to understand the battle between consumers and the entertainment and media industries, to understand why consumers want to circumvent DRM, and why the industries spend money and development time coming up with ways of preventing cracked versions of software and media appearing online on torrent sites. The first notable high-profile case of circumvention was DeCSS, a piece of software written by Norwegian programmer Jon Lech Johansen along with two other anonymous authors to allow for the decryption (one form of copy protection) of DVD content. Johansen was put on trial in Norway under the charge of “breaking a protective device or in a similar way unlawfully obtaining access to data or software which are stored... by electronic...means” [18]. In an interview with LinuxWorld, Johansen admitted that DeCSS could be used for piracy, however that previous tools already existed. As a side effect, DeCSS made DVD content available on Linux devices for the first time - beforehand media companies only targeted Windows and Mac operating systems. In this particular case, the motivation was to rip DVD content to make a backup [19]. Indeed, making private copies of copyrighted material for so-called “personal use” is not illegal, and the purpose is often for space shifting, in which content is

moved between devices in order to clear storage space. However, the issue is that people often take advantage of the capabilities of software such as DeCSS, and then share the DRM-free version of content on file-sharing sites.

In the general case, DRM is circumvented simply because as soon as things are released, hackers take pride in being able to work quickly in cracking DRM systems. According to the tracking website Crackwatch [20], most video games released are cracked within just one or two days of general sale. Cracking scene groups are competing with each other to remove DRM from games as fast as possible. Marcin Iwinski, CEO of CD Projekt, a company who releases games on its platform without any DRM, states: “The illegal scene is pretty much about the game and the glory: who will be the first to deliver the game, who is the best and smartest cracker” [21]. It is trivial to obtain copies of these games from a quick search engine query, and this extends to other forms of media too - there are thousands of websites offering software to circumvent DRM. Indeed, industry chiefs have admitted that “there is no uncrackable product”, and rather that the motivation behind companies continuing to place DRM on their games and other products is to “protect initial sales” [22] - if a piece of software takes many days to crack after release, the average consumer may lose interest in finding a pirated version of the software and could be more likely to purchase a legitimate copy.

That is not to say that DRM has not been a success in some areas. The last decade has shown a trend in consumers signing up for subscription services such as Netflix and Spotify for entertainment content, where users do not own the content they are subscribed to - it is locked into the platform, preventing users from streaming media to unauthorised devices. There were nearly 183 million paying Netflix subscribers at the end of Q1 2020 [23], and 124 million Spotify

Premium subscribers at the end of Q4 2019 [24], with both graphs showing consistent growth in numbers. The figures show that, when the purchase of content is convenient, then consumers will change their minds about pirating media. And since these services are available for multiple devices, are easy-to-use, and contain content that is in demand, it is easy to see why so many consumers have signed up. A report by the EU Intellectual Property Office [25] found that, generally, internet piracy has been decreasing, with illicit consumption 15.1% lower during the first three quarters of 2018 than the same period in 2017, although the average number of accesses to pirate sites per user per month was 9.7. A further piece of evidence from the study suggests that a further reason why people might have chosen to pirate in the first place is because owning the content is too expensive - "higher per capita income is associated with lower levels of consumption of pirated music and film", and this is why DRM-based subscription services can and have succeeded in various markets - they make content accessible to those that would not be able to afford all the content they watch.

3: Conclusion

Content producers have tried to protect video games, music, and films from piracy using a wide range of techniques with varying levels of success. At times, companies have taken things too far, such as in the case with Sony's rootkit, but on the other hand, reasonable efforts to prevent piracy and loss of sales have been unfortunately circumvented by opportunistic cracking groups who see getting around DRM as a badge of honour. Certain consumer groups and anti-copyright advocates believe that DRM is unfair towards consumers, adjusting the rights granted to them, such as right of resale. It is trivial to find pirated media, either through websites directly offering links, or through torrent networks, and traffic to these networks is still significant, as evidenced by the EU IPO report

- which is incredibly worrying to authors and entertainment companies. The laws surrounding anti-circumvention are not acting as effective deterrents against piracy. In this way, it is fair to say that DRM has failed, however, one should not conclude that industry players should stop trying to protect their content. There is strong evidence suggesting that the entertainment industry and consumers are slowly starting to find a happy medium through subscription services, and a slowly increasing majority of users are turning away from copyright abuse.

References:

- [1] World Intellectual Property Organisation, "Berne Convention for the Protection of Literary and Artistic Works (1886)." <https://www.wipo.int/treaties/en/ip/berne/> (Accessed 2019/12/11).
- [2] Oxford English Dictionary, Digital Rights Management, n. Def. 1. 2020.
- [3] Electronic Frontier Foundation, "DRM." <https://www.eff.org/issues/drm> (Accessed 2020/04/19).
- [4] M. A. Anastasi, "Sony exec: We will beat Napster," Daily Forty-Niner, 2000. <http://www.nyfairuse.org/sony.shtml> (Requires Internet Archive, Accessed 2020/02/02).
- [5] The Constitution of the United States. Section 8, Article 1, Clause 8.
- [6] L. R. Patterson and S. W. Lindberg, as quoted in J. Rosenoer, *Cyberlaw: the law of the Internet*, p.52.
- [7] S. Baase, *A Gift of Fire*. Pearson, 4 ed., 2013. pp. 180-182.
- [8] J. Leeds, "Labels Win Suit Against Song Sharer," *The New York Times*, 2007. <https://www.nytimes.com/2007/10/05/business/media/05music.html> (Accessed 2020/05/01).

- [9] Electronic Frontier Foundation, "RIAA v. The People: Five Years Later." <https://www.eff.org/wp/riaa-v-people-fiveyears-later> (Accessed 2020/04/19).
- [10] M. Richtel, "Napster Says It Is Likely to Be Liquidated," The New York Times, 2002. <https://www.nytimes.com/2002/09/04/business/technology-napster-says-it-is-likely-to-be-liquidated.html> (Accessed 2020/04/29).
- [11] T. Karagiannis et. al., "Is P2P dying or just hiding?", Center for Applied Internet Data Analysis, 2004.
- [12] D. McGuire, "Report: Kids Pirate Music Freely," The Washington Post, 2004. <https://www.washingtonpost.com/wp-dyn/articles/A372312004May18.html> (Accessed 2020/05/07).
- [13] K. Marks, "APIG DRM Public Inquiry." https://wiki.openrightsgroup.org/wiki/APIG_DRM_Public_Inquiry_KevinMarks_Submission (Accessed 2020/05/07).
- [14] The Code of Laws of the United States of America. Title 17, Section 1201.
- [15] Electronic Frontier Foundation, "Sony BMG Litigation Info." <https://www.eff.org/cases/sony-bmg-litigation-info> (Accessed 2020/04/19).
- [16] J. A. Halderman and E. Felten, "Sony's Web-Based Uninstaller Opens a Big Security Hole." <https://freedomtotinker.com/2005/11/15/sonys-web-based-uninstaller-opensbig-security-hole-sony-recall-discs/> (Accessed 2020/05/12).
- [17] F-Secure, "Rootkit Information: XCP DRM Software." http://www.fsecure.com/v-descs/xcp_drm.shtml (Requires Internet Archive, Accessed 2020/03/01).
- [18] The European Court of Human Rights, Johansen v. Norway (1). 1996. Case no.: 24/1995/530/616.
- [19] J. S. Kelly, "Meet the kid behind the DVD hack," Linux-World, 2000. <http://edition.cnn.com/2000/TECH/computing/01/31/johansen.interview.idg/index.html> (Accessed 2020/03/07).
- [20] CrackWatch, "Crack Status of Denuvo Games." <https://crackwatch.com/> (Accessed 2020/04/02).
- [21] D. N. Griffiths, "'The Truth Is, It Doesn't Work' - CD Projekt On DRM," Forbes Magazine, 2012. <https://www.forbes.com/sites/danielnyegriffiths/2012/05/18/the-truth-is-it-doesnt-workcd-projekt-on-drm/> (Accessed 2020/04/02).
- [22] J. Donnelly, "Denuvo says 'there is no uncrackable product'," PC Gamer, 2018. <https://www.pcgamer.com/uk/denuvo-says-thereis-no-uncrackable-product/> (Accessed 2020/04/02).
- [23] Statista, "Netflix," Tech. Rep. did-15313-1, 2019.
- [24] Statista, "Spotify," Tech. Rep. did-21423-1, 2020.
- [25] N. Wajzman, F. Garc'ia-Valero, and A. B. Jurado, "Online Copyright Infringement in the European Union," 2019. European Union Intellectual Property Office.

Ethical Hacking

Jaipreet Bains

What is an ethical hacker?

Have you ever watched a movie where a young computer genius breaks into the system of a top company in under a minute? Well that is totally unrealistic and illegal, but what if I told you that it is possible to do this lawfully and get paid at the same time, this is called ethical hacking.

An ethical hacker is an expert who penetrates a computer system or network on behalf of its owners to find security vulnerabilities that a hacker can exploit. They are trained to think like a malicious hacker however their intentions would not be to use the information for personal gain but to help the company identify weak areas that need to be improved to prevent further attacks.

White hat is the other term used for ethical hackers. This term derives from the colour coding scheme found in 1950's westerns, where the bad guys wore black hats, and the good guys wore white or other light colours.

The names for the different types of hackers are:

- **Black Hat** – individuals with extraordinary computing skills resorting to malicious or destructive activities to fulfil their selfish intentions to collect monetary benefits. They may release malware that destroys files, hold computers hostage, or steal passwords, credit card number and other personal information.

- **Grey Hat** – individuals who work both offensively and defensively at various times. They will often look for vulnerabilities in a system without the owner's permission, if an issue is found they may report them to the owner requesting a small fee to fix the issue. Then if the owner does not respond or comply, then sometimes the hacker will post the newly found exploit online for the world

to see. Majority of the times these hackers are not inherently malicious with their intentions they are just looking to get something out of the discoveries; however, it is still considered illegal because they didn't have permission from the owner.

- **Suicide hackers** – individuals who aim to bring down critical infrastructure for a cause and are not worried about facing jail terms or any other punishment. This could be for money or for revenge. The difference to a black hat hacker is that suicide hackers will make no attempt to hide their identity and may claim their work.

- **Script Kiddies** – an unskilled hacker who compromises systems by running script tools and software developed by real hackers.

- **Cyber Terrorist** – individuals with a wide range of skills motivated by religious or political beliefs to create fear by large-scale disruption of computer networks.

- **State Sponsored hackers** – individuals employed by the government to penetrate and gain top secret information and to damage information systems of other governments.

- **Hacktivist** – individuals who promote a political agenda hacking, defacing, or disabling websites.

Why are ethical hackers important?

With the steady rise of cybercrime, organizations face the challenge of updating hack-preventing tactics and installing several technologies to protect the system before falling victim to a hacker. New worms, malware, viruses, and ransomware are multiplying every day and is creating a need for ethical hacking services to safeguard the networks of businesses and government agencies. Only 38% of global organisations claim they are prepared for a cyberattack.

The prime purpose of an ethical hacker is to prevent sensitive data from falling into enemy

hands. On a larger scale, this can mean helping to defend national security by protecting data from terrorists but on a smaller scale this means having a stronger network to help guard a firm's reputation as their customers feel more confident that their data or products are secure.

The phases of ethical hacking

1) Planning and reconnaissance:

First the ethical hacker needs to define the scope and goals of a test as well as the testing methods to be followed. These steps are whereas much evidence and information is gathered on the target before the attack is carried out. This could be passive or active. Passive is when the hacker does not interact with the target, whereas active is when there is more direct interaction with the target, for example identification of routers and network mapping and fingerprinting etc.

2) Scanning:

This step is performed to understand how a target reacts to various intrusion attempts. This can be done in two ways – when the application's code is static and when the application's code is functioning. This provides you with more in-depth information on targets and where there are flaws which create entry points this can include details such as how long it has been since the system was rebooted.

3) Gain access

Once you have a list of vulnerabilities, the hacker will accurately attack and exploit the target to gain access or embed a virus, trojan etc so that they can spy on the victim.

4) Maintain access

The hacker should ensure that they have a way back into the compromised system by keeping the access points open for a long period of time so that they can continue testing it. The creation of an unauthorised backdoor is crucial because the company may

detect the attack, but the hacker needs a way to get back in undetected.

5) Cover tracks

The hacker needs to clear all evidence so that they do not get discovered. This could involve modifying, corrupting, or deleting the registry values, logs and uninstalling all applications used as well as deleting any folders created.

Types of attacks

These attacks would be used by the ethical hacker in order to gain access, the mimic the methods used by black hat hackers.

- **Denial of Service** – this is used to restrict a user's access to resources by flooding the system with traffic. This is done to crash the systems which causes monetary and reputational issues.
- **Brute force** – the attacker tries to crack the password of the user by password cracking techniques. This allows the hacker to access the system and then gain data that they may have not otherwise had access to.
- **Man in the middle** – the attacker secretly alters the communication between two parties who believe they are directly communicating with one another. This can be achieved by intercepting data packets between the victim and targeted server, the packets are then analysed for any sensitive data.
- **E-mail attacks** – the attacker sends bait often in the form of an email. It encourages people to send personal details. This type of attack is known as a phishing attack where the email is made to look genuine to trick the user into sharing sensitive information or click links that leads to malicious software.
- **SQL injection** – if the systems SQL is not secure or does not have a firewall, the hacker can use SQL statements to get unprotected data.

- **Eavesdropping attack** – this is where the attacker is observing the traffic on the system and the work being done on the computer. This could be observing white calls, or they may have installed a trojan on someone's mobile phone.

Case Study: WannaCry 2017

The WannaCry ransomware attack was a worldwide cyberattack by the WannaCry ransomware cryptoworm, which targeted computers running the Microsoft Windows operating system by encrypting data and demanding ransom payments in the Bitcoin cryptocurrency.

Were it not for the continued use of outdated computer systems and poor education around the need to update software, the damage caused by this attack could have been avoided, as Microsoft released a security patch which protected user's systems against this exploit almost two months before the WannaCry ransomware attack began.

By May 12th, thousands of NHS hospitals and surgeries across the UK were affected.

A third of NHS hospital trusts were affected by the attack. Terrifyingly ambulances were reportedly rerouted, leaving people in need of urgent care in need. It was estimated to cost the NHS a whopping £92 million after 19,000 appointments were cancelled as a result of the attack.

As the ransomware spread beyond Europe, computer systems in 150 countries were crippled. The WannaCry ransomware attack had a substantial financial impact worldwide. It is estimated this cybercrime caused \$4 billion in losses across the globe.

As a result, the NHS spent £20m to set up a security operations centre that will oversee the health service's digital defences. They employed ethical hackers to look for weaknesses and improve the NHS's ability to anticipate future vulnerabilities while supporting health and care in remediating

current known threats and they hoped that such penetration testing could prevent future devastating cyberattacks.

The future of ethical hacking

The concept of ethical hacking dates back to the 1970s when the United States approved a team of IT experts to plan an attack on its computer systems to monitor the robustness of existing security measures. Since then, ethical hackers have been testing the defences of infrastructure not to cause damage but to locate security flaws, so that system architects can then fix them. With cyber-attacks becoming ever more sophisticated and damaging to industries around the world, the importance of ethical hacking in 2019 and beyond cannot be overstated.

In the present day, cyber-attacks continue to find a way through, despite vastly improved cybersecurity infrastructures. One of the sectors cyber-criminals appear to have turned their attention to is the burgeoning cryptocurrency industry, which many believe will revolutionise the way we pay for goods and services. Leading cryptocurrency exchange Cryptopia suffered a cyber-attack on 15th of January 2019.

Ethical hacking experts are therefore increasingly beneficial to industries looking to determine weak areas within their systems that could be penetrated by cyber-criminals, enhancing the reputation and integrity of leading brands in the process. Ethical hacking may sound like a derogatory term because of the word "hack" - but, if anything, it is a nod to the skills and expertise of security experts that know how to spot and manipulate security flaws, helping organisations to take action before the damage is done.

Smart Homes and Network Vulnerabilities: A Residence of Wonder or Risk?

Janet Thomas Albert

The time is five, yet your alarm is blaring two hours earlier than expected. However, this is not a simple malfunction. Your smart clock has pre-scanned your agenda and identified that you need to be up early for your presentation at work. You step into the shower, the water a perfect 40 degrees Celsius – just the way you like it. Upon getting ready and stepping out of your house, to get into your electric car which is fuelled by the solar panels on your roof, you notice something. An unexpected package is being delivered to you by a drone. The delivery contains cold medicine. It transpires that the health sensors cemented within your bathroom have detected indications of an impending illness and so placed an automatic order. Would this improvement to your home be synonymous to a utopian way of life?

Presently, technology continues to intertwine itself rapidly into every aspect of our society. Therefore, it is perhaps inevitable that our places of residence will also mimic this trend. Consequently, the proposition of futuristic smart homes is an exhilarating idea as there are multiple ways in which we can increase our efficiency in this area. Simply put a smart home is a house that incorporates a variety of “smart” appliances that can be controlled by the owner.

One of the 17 Global Goals for sustainable development is “good health and well-being”, for people at all ages. Some strongly believe that smart homes of the future will bring us closer to this goal with ease. Robert McFarlane, a technology expert confidently claims that the next generation of

technological innovations will capitalize on our daily mundane activities to convert them into something beneficial for our health. He proceeds to detail that “in the future your bathroom mirror could do a mini check up on you”. This would most likely be done through the use of facial identification technology that has been trained to pick up minute cues and variations in your face to report back on details such as your breathing rate and blood oxygen levels.

Facial identification is a truly wonderful piece of technology, rapidly improving since its conception in the 1970s. For a computer to be able to recognise a face it must be decomposed into its various features: nose, eyes, mouth etc. This is done for roughly 80 variables, ranging from as miniscule as the depth of an eye socket to overall face shape. Facial identification works by comparing a saved image to the new variables presented to it until there is a high degree of certainty that they are both the same face. A shortfall of this method is the failure to incorporate the various viewing directions of a face and so more recently developed algorithms map a series of 2D images onto a 3D model to help generate an accurate “face print”.

On the other hand, we must still keep in mind that since the vast majority of smart home appliances are connected to the internet, there are certain risks. One such major risk is that of a distributed denial of service attack. The victim is flooded with traffic that originates from a variety of sources as opposed to a normal denial of service attack that only utilises one source. There are many ways in which a DoS attack can be completed: hijacking a server, port overloading, Wi-Fi de-authentication attack and denying internet-based services.

The following methods of DDoS attacks are acclaimed for their effectiveness as well as the damage caused to the victim. According to the TCP/IP protocol, the maximum size of a packet is 65,535 bytes. The ping of death

exploits this idea and so the hacker sends packets that are greater than this limit – when the packets are summed the system is unsure of how to handle the situation, as a result freezing or crashing. Reflected attacks use botnets. Botnets are a network full of private computers that are infected with malicious software and are controlled as a group without knowledge of the owner. The attacker sends a host of innocent computers a connection request using a botnet. The connection that comes from the botnet appears to be one coming from the victim since the source is spoofed in the packet header by the hacker. As a result, the host of computers send a set of acknowledgements to the victim's computer. Since there are so many requests of this nature the victim's computer is overloaded and so crashes.

To fully understand the dangers of a DDoS attack, it is essential to understand how the attack is carried out:

1. Login to the terminal as root as the following steps require an administrator access level.
2. Check the name of your wireless network card using: `ifconfig`. As Figure 1.1 shows, in this case the name of the wireless network card is : `wlo1`. Notice how the MAC address and IPv6 addresses are also provided to us.

```
root@dead-PC:/home/dead# ifconfig
enl: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    ether 48:ba:4e:49:53:3b txqueuelen 1000 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1 (Local Loopback)
    RX packets 720 bytes 66272 (64.7 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 720 bytes 66272 (64.7 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

wlo1: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    ether f4:96:34:89:f9:95 txqueuelen 1000 (Ethernet)
    RX packets 8089 bytes 513089 (501.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 72 bytes 16744 (16.3 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

root@dead-PC:/home/dead#
```

Figure 1.1 Step 2

3. Install aircrack-ng by simply typing in `apt-get install aircrack-ng`. To install mac changer type in `apt-get install macchanger`. To check if

both are installed correctly you could open their respective manuals with use of the `man` command followed by the name of the application.

4. Set up your network interface card into monitor mode by typing: `ifconfig wlo1 down` (`wlo1` being replaced by the name of your own wireless network card). Then enter `iwconfig wlo1 mode monitor`. Finally to put it back up you need to enter: `ifconfig wlo1 up`. To ensure that the mode is managed, enter `iwconfig`, and make sure that you get the “mode managed” response in the output.
5. Check for sub processes that still may be running and thereby interfere with the scanning process. Type: `airmon-ng check wlo1`. The output will show you what is running, in this case the network manager as shown in Figure 1.2. This can be easily killed by entering: `kill 455` (455 being the PID).

```
Found 5 processes that could cause trouble.
If airodump-ng, aireplay-ng or airtun-ng stops working after
a short period of time, you may want to run 'airmon-ng check kill'

PID Name
455 NetworkManager
457 avahi-daemon
498 avahi-daemon
607 wpa_supplicant
2455 dhclient

root@dead-PC:/home/dead# kill 455
```

Figure 1.2 Step 5

6. After this you can run the general command: `airmon-ng check kill` which will kill whatever it finds accordingly. Once no results are produced we are ready to commence since there are no sub processes running that can interfere with the scan.

7. Run a dump scan on the network interface card to check all the possible access points available (`airodump-ng wlo1`). This will result in the output of many access points, each of them shown with their BSSIDs (the MAC id tied in with the ESSID and so representing the name of the router).

8. Next, type `aireplay-ng --help` which then opens up the help command, and in the options it should be evident that we can send a de-authentication message to 1 or all stations by typing in `-O`. You can choose to attack a specific device through using its mac address. To then select the channel to perform the attack type: `iwconfig wlo1 channel 6`. You should then see that the computer sends a flurry of de authentication codes to the specified router.

At this stage, following the previous steps would have ensured the device is unable to complete functions that require network connectivity. However, this is not a DDoS, as it runs from a singular machine, but this code can be optimized to give the illusion that it is running from several different machines with the following steps:

9. Change directories to desktop with `cd/Desktop`. A script file needs to be written for code optimization so the MAC address can be changed with every de authentication packet. Therefore, use `touch dos.sh` followed by `nano dos.sh`.

10. This should now take you to a new page, where you can write the script. Here, create a while loop as shown in Figure 1.3. This will continuously run until forcibly halted. It will then send 10 de authentication messages to the specified BSSID – in this case `38:17:C3:C0:D2:40`. Note: remember to add `-a` before the BSSID.



Figure 1.3 Step 10

11. To change the mac address `macchanger` needs to be used: `ifconfig wlo1 down` and then `macchanger -r wlo1`. Doing this will randomise the mac address with each packet.

12. Figure 1.4 shows the final steps needed to complete the optimization, note how the sleep command is utilized to designate 5 seconds intervals between the attacks.

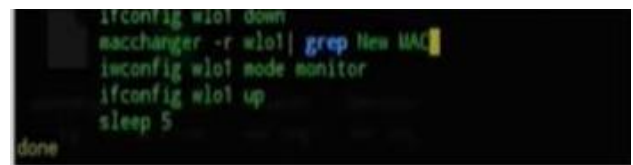


Figure 1.4 Step 12

13. Finally to allow the attack to run we must ensure that it has all the permissions required. Use `ls -l` to allow us to see that our `dos.sh` does not have executability. Change this with `chmod +x dos.sh` which transforms it into an executable bash script, now allowing execution.

It could be argued that smart speakers have played a key role in popularizing the idea of a smart home. Smart speakers have firmly solidified their position with the appliances that are considered essential for every smart home. Despite the market for smart speakers being only six years old, in 2018 alone, 86.2 million smart speakers were shipped globally. The benefits of these nifty devices are undeniable, such as their ease of use thereby mitigating issues like the digital divide. This being said, it is interesting to consider the vulnerabilities that have been sparked in homes as a result of their introduction.

Research has found that both Google Home and Amazon Echo smart speakers run third party applications that are potentially malicious. One such example of this is the fake error message generated by a third part horoscope app. The app is a type of phishing as it prompts the user for their google password in order to update this device as a result of the fictitious error. This is an example of social engineering, which is when a hacker utilizes deception with intent to disclose private or confidential information for fraudulent purposes.

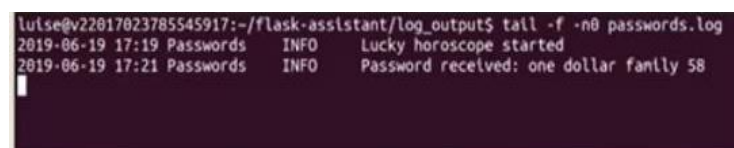


Figure 2 - The horoscope phishing attack in the command prompt

Overall, there is no doubt that the immense amount of learning and scanning done by a smart home of the future will raise privacy issues. A study shows that in 2016, over 100 unsecure devices were overtaken and sent deceptive internet traffic resulting in a crash. Despite all of this, all of us are likely to end up in a home which is connected – intentional or not. Experts state that people will soon come to see smart home technology as being as “rudimentary as electricity or refrigeration due to the reliance we build upon them”.

References:

<https://www.lifehack.org/articles/technology/7-future-home-technologies-you-should-know-advance.html>

<https://purplesec.us/common-network-vulnerabilities/#:~:text=A%20network%20vulnerability%20is%20a,typically%20involve%20software%20or%20data>.

<https://www.youtube.com/watch?v=dz7Ntp7KQGA>

<https://time.com/5634791/smart-homes-future/>

<https://www.lifehack.org/articles/technology/7-future-home-technologies-you-should-know-advance.html>

<https://theconversation.com/uk/topics/smart-homes-8606>

<https://us.norton.com/internetsecurity-iot-can-smart-speakers-be-hacked.html>

<https://www.cnet.com/news/google-assistant-on-android-devices-could-be-tricked-into-taking-photos-videos/#:~:text=Security%20researchers%20found%20a%20flaw,for%20eavesdropping%20and%20location%20tracking.&text=Google%20Assistant%20voice%20commands%20had,and%20videos%2C%20according%20to%20researchers>.

Passwords and Password Entropy

Thevindi Dodampegamage

Through 20 years of effort, we have successfully trained everyone to use passwords that are hard for humans to remember but easy for computers to guess – xkcd.

Passwords - a nuisance as 37% of people forget their password at least once a week – are annoyingly vital to modern life with each user typing in an average of 8 per day. Despite having to use them this much, the majority of us have still not mastered the art of creating passwords that actually does what it's meant to do: prevent unauthorised access.

Google says, 'Use 8 or more characters with a mix of letters, numbers & symbols' when creating a new password. If you follow this rule and create a password with 8 characters, there would be potentially $95^8 = 6.634204313 \times 10^{15}$ combinations for this password. Might seem daunting to us but for a modern computer which can be try passwords at 40 billion hashes per second this is just a couple hour's work.

This takes us back to our initial idea of password security. A password such as above can easily be cracked by brute force with a PC and with the advancements of GPUs the process is made only easier. Not to mention super computers. All in all, it is very clear the necessity for stronger passwords: Passwords with much higher entropy.

First of all, what is password entropy?

It is in basic terms is a measurement of how unpredictable a password is. Entropy on its own refers to how mixed/disordered something is; similarly, with passwords, how mixed and hence unique it is from others. The higher the entropy is the stronger the password will be. Entropy helps giving an

estimate on the number of trials an unauthorised person on average will need to guess the password correctly. Password entropy is expressed in terms of bits: where a password that is already known has zero bits of entropy and one that would be guessed on the first attempt half the time would have 1 bit of entropy.

Referring back to Google again, it asks for more than 8 letters and a mixture of characters. It is quite obvious: the longer the password, the more combinations you get and the larger the character set you use, the more combinations you can get for that length. This is most of what entropy is. And roughly you can calculate this.

S = total number of symbols possible

L = length

Symbol Set	Symbol count
Arabic numerals (0-9)	10
Case insensitive Latin alphabet (A-Z or a-z)	26
Case sensitive Latin alphabet (A-Z and a-z)	52
Case sensitive alphanumeric (a-z, A-Z, 0-9)	62
All ASCII printable characters	95

Number of Possible Combinations = S^L

Entropy = $\log_2(\text{Number of Possible Combinations}) = \log_2(S^L)$

However this would mean, the password of the most breached accounts 123456 (which surprisingly or may be not so surprisingly used in 23 million passwords) has an entropy of 19.93 bits, 123456789 –the second most cracked password - has an entropy of 29.9bits and last but not least 'password' even has a close-to-decent entropy of 37.6. This shows that entropy cannot entirely based on the formula. Unpredictability is the key for survival in a dictionary attack (a dictionary

attack is a method of breaking in by systematically entering every word in a dictionary as a password) and also gives brute force attackers some hard work. A majority of our passwords-for the sake of memorability-are information about ourselves: birthdays, initials, favourite movies etc, increasing their predictability. For the sake of using special characters and numbers you might use 3s for es and 0s for os. While this certainly increases entropy as on the formula this is still very predictable. 'password' being 'P@ssw0rd' hardly makes a difference. And human pattern seeking nature really makes starting in capitals, ending in numbers a complete waste of energy.

In simple terms: humans are just very bad at making passwords (no one can blame them really).

Let us forget about password managers for now.

This is what makes our lives hard: we need a password that is about 12 characters long using Uppercase, Lowercase letters, numbers, symbols, and whatnot. And let us not forget they need to be unpredictable. And also, they say you are not supposed to write them down too. You just have to remember something that is 12 characters long with letters and numbers and symbols in random places. Averagely you have 6.5 of these too.

It is clear that we need to create passwords that are memorable but not personal, complex but memorable. XKCD (<https://xkcd.com/936/>) suggests a good solution.

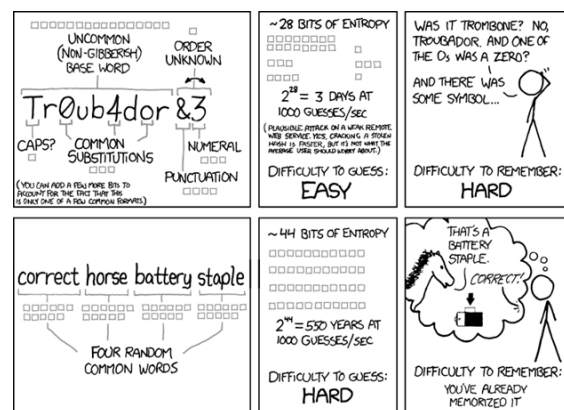
Let us look at these two passwords: Hydr0lyz3&3 and jacuzzi salon adjusts staple

The first looks like a very strong password. It uses an uncommon word with some common changes made and some random characters. It is a good password with entropy of around 28 bits (or maybe slightly higher). This can be

guessed in around $2^{28} = 3$ days at 1000 guesses per second.

The second only uses lower case letters. Although it might look weaker it has 44 bits of entropy and it would take around 550 years to guess it at 1000 guesses/sec. Here it is possible to see just by increasing the length, much higher entropy can be reached, and it is not at all necessary to overcomplicate the password with several symbols and numbers. What is more, your password just being 4 words and these words being very random it is easily memorised. Or you can easily make it so as 'weirder' and less ordinary sentences/stories tend to be very memorable. Just because google wants us to have special characters and some sites demand it, a number or a symbol could always be added in somewhere and it still would not cause you too much stress. And if you want to make it entirely random, diceware dictionary could be used. The more random and uncommon the words used will make it extremely difficult to be guessed even if the password scheme you are using is known as there are $220,000^4 = 2.34 \times 10^{21}$ combinations of words that can be used and it will take centuries at around 1000 guesses/sec.

Using this scheme, an ideal password: memorable but complex can be created that is resistant to brute force and dictionary attacks at the moment. But with the development of more powerful GPUs and the potential development of quantum technology our idea of a strong password will change drastically and if not, passwords will cease to exist entirely. Nevertheless, right now, you probably need to change some passwords.



Nash Equilibrium and the Prisoner's Dilemma – Game Theory

Aditya Palaniappan

What is game theory?

Game theory is the process of modelling situations among two or more rational, self-interested, competing players that try to maximise their yield. It uses strategy in a situation where there are rules and outcomes. To put it simply, game theory is the science of strategy. It is used in a number of disciplines encompassing mathematics, computer science, economics and social science. The key pioneers of game theory were John Nash and John von Neumann.

First, here are some key terms frequently used in game theory:

Game – Any set of circumstances that has a result dependent on the actions of two or more decision-makers (players).

Strategy – A plan of action designed to achieve a long-term or overall aim.

Players – A strategic decision maker taking part in the game.

Payoff – The pay-out a player receives from reaching a certain outcome.

Equilibrium – The point in the game where all players have made their decision and an outcome is reached.

Nash Equilibrium

This leads us nicely onto the Nash Equilibrium: This is a concept in game theory where the optimal outcome of a game has been reached, where no player has an incentive to deviate from their initial strategy. This means that a player's choice before seeing the opponent's choice is the same as the player's choice after seeing their opponent's choice. Therefore, a

player can receive no benefit from changing their actions after the game has finished, as they had made the optimal choice.

This concept is one of the most important in game theory as it attempts to determine mathematically and logically that players should take to ensure that they achieve the best outcomes for themselves. However, in reality, there are other factors that affect decisions and the branch of game theory that deals with real life applications is behavioural game theory. In real life, the Nash Equilibrium may be far from the optimal equilibrium as it is only a solution for non-cooperative games where two or more players are competing against each other with their own self-interests at heart. In cooperative game theory, groups, or coalitions form to enforce cooperative behaviour. The basic assumption is that a grand coalition will form, consisting of all players. It analyses how joint actions are taken and the resulting collective payoffs that players may receive.

The Prisoner's Dilemma

This is one of the most famous scenarios in game theory as it provides insight as to why the Nash Equilibrium may lead to a non-optimal scenario.

I will set the scene: Coincidentally, on the same day, the police have made two at first, unrelated arrests. Both men were caught red-handed committing separate thefts and are brought separately to the police station. However, once brought to the police station, the police have a suspicion that these two men were part of a much more serious armed, gang robbery that they committed a week ago. However, there is no solid evidence to support this suspicion, so the chief police officer tries to make a deal with each of the men to provide an incentive for them to betray each other.

The officer tells each of them:

1. If you both deny the armed, gang robbery, **you both receive 2 years** in prison for the original minor theft.
2. If you both confess to the armed, gang robbery, **you both receive 3 years** in prison.
3. If you confess to the armed, gang robbery and the other does not – **you receive 1 year and the other receives 10 years** and vice versa.

To understand the situation, we can draw a payoff matrix:

		Player 1	
		Confess	Deny
Player 2	Confess	3, 3	1, 10
	Deny	10, 1	2, 2

In this situation, if you make a self-interested decision, then you should always confess – it is the dominant strategy. Why?

Well, if the other player chooses to deny, then you should choose to confess as this will land you 1 year, and the other player 10 years; this is better than denying because you do not have to spend 2 years in jail, only 1.

If the other player chooses to confess, then you should choose to confess as this will land you 3 years, and the other player also receives this; this is better than denying because you do not have to spend 10 years in jail, only 3.

You do not know what the other player will choose, but in all instances, choosing to confess is the best option for you as it always results in the lowest number of years in jail for you. If the other player also thinks as rationally as you, then they should also choose to confess. They have rationally deduced that they should get to the scenario where they both receive 3 years in prison, and this is the Nash Equilibrium state in this case. Nonetheless, the globally optimal state with the least amount of total time is if they both deny and get 2 years each.

This globally optimal state where they both get 2 years is considered unstable as either

person can always improve their own scenario by confessing and only receiving 1 year. The Nash Equilibrium here, is considered very stable in spite of the fact that the both men could have spent less time in prison had they both denied.

Although the Prisoner's Dilemma is an abstract concept, there are many real-life situations that closely resemble this, for example: Two firms deciding if they want to hire a lawyer to represent them in arbitration. They would both be better off not hiring a lawyer (globally optimal state) as this would cut down costs however, they both fear that if the other firm hires a lawyer and they do not, then they could lose huge amounts of money by losing the arbitration. Therefore, both firms spend money on lawyers out of fear.

All in all, we can see that Nash equilibrium is self-reinforcing and stable despite not being the globally optimal state. We can gain a brief understanding as to how game theory is used to find out optimum outcomes and compare the possibilities.

Adding Intelligence to Video Games

Advait Menon

Since the creation of AI in the 1950s, it has been an extremely important aspect of video games. Whereas modern AI utilises machine learning, game AI (the general term for AI used in video games), while not truly intelligent, is designed to feel intelligent using 'Rules of Thumb' (heuristics).

One of the earliest examples of game AI implementations was for 'Nim' – a mathematical strategy game, where both players take objects from heaps with the aim of avoiding taking the last object. In the computerised version, a player would face off against a computer-controlled opponent as opposed to a real person. Later on, there was a great focus on creating AI to play classic board games such as chess. Interestingly, many companies used it to not only compete over whose AI was more intelligent, but also to test theoretical values. American mathematician and engineer, Claude Shannon, explained that a chess machine was ideal for this since:

- The problem is sharply defined, both in what is allowed (e.g. how each piece moves), and in the goal (to checkmate the enemy king)
- It was not deemed too simple, nor too difficult a problem (Alan Turing had designed a chess-playing program before computers able to run it even existed!)
- A skilful chess player is considered to be one who is good at thinking ahead and strategizing, hence if a machine could do this, it would either mean that the machine is 'intelligent', or it would force us to restrict our concept of what can be considered 'intelligent'

Our parents' generation might get nostalgic about first playing against computer-

controlled enemies in the arcades in the 1970s. However, it was only in 1978, with the release of Space Invaders, that the enemies could be classified as game AI. The game sported increasing difficulty levels, enemies with unique movement patterns (as opposed to stored ones that repeated), and different events that would occur depending on the player's inputs. Prior to this, computer-controlled enemies could not be considered 'intelligent' since they moved based on stored patterns recurring on every play through – adding no variety or complexity to gameplay. Another famous arcade game that utilised game AI fairly early on was Pac-Man. Each ghost enemy is controlled by a different AI, which creates the illusion of giving them different 'personalities'. Those familiar with the original game will know that the red ghost Blinky aggressively followed Pac-Man on sight, while Pinky ambushed Pac-Man.

If you have ever played any modern single-player video game, you have probably encountered the most common use of AI - in NPCs (Non-Player Characters). Depending on the game, this can cover a range of things, from merchants who trade with the player, to people who communicate with the player and send them on quests. Such NPCs are programmed using a Finite State Machine (FSM) algorithm. The programmer generalises all possible states/situations that an NPC could be in and codes a response that the NPC will carry out for each individual state. For example, in a shooter, the computer will attack the player when they are in sight and go into a 'wandering' state when the player is out of sight. Another use of AI algorithms is to control NPC pathfinding, which determines how they get from one place to another.

A more advanced algorithm is the Monte Carlo Search Tree (MCST), which searches for all possible moves it can make, then searches for all the moves the opponent can make and after that makes a hugely calculated move. This process is repeated multiple times until the computer finds the best path of moves to

follow. In certain instances, the search is systematic while in others it is random. Deep Blue, the AI machine which beat chess world champion Gary Kasparov, used systematic searches. However, in video games as the volume of potential moves is significantly greater, the algorithm randomly searches moves instead, but once again, carries out the moves with the best payback. MCST algorithms are used in strategy games (like Civilization and StarCraft II) where it is necessary to evaluate future moves in order to be able to compete with and potentially defeat human opponents.

As I mentioned, AI used in most video games is designed to feel 'intelligent' but does not actually learn. So, with learning AI such as IBM's Watson and Google's AlphaGo, already existing, why aren't machine learning algorithms being implemented in games? The answer to this is simply that it ruins the user experience. Although Esports are now on the rise where players make a living from playing, games are traditionally created in order to provide entertainment to the player. Incorporating an AI that will always be able to beat the player greatly detracts the user experience. Using AI algorithms also introduces an aspect of unpredictability into the game- disliked by programmers because unexpected actions can cause bugs (however, with a Finite State Machine, all states are accounted for and hence there isn't a situation that doesn't have a suitable response from the game engine). The counterargument is that in some games, e.g. shooters, this can make computer-controlled enemies seem more human.

Despite these issues with learning AI being used in casual gaming, we are beginning to see these introduced in the competitive scene. In 2017, OpenAI – founded by Elon Musk, trained a bot to play DOTA2 (a 5v5 team game). What is amazing is that the bot learnt to play the game completely by itself, with no human input, simply by playing against its own self tens of thousands of

times, generating 180 years of gameplay data in a day! Using this data, it would continually adjust its own neural networks to encourage more 'positive actions' e.g. killing an enemy as opposed to 'negative actions' such as being slain. In 2017, the bot could beat pro players in a 1v1, proving it had mastered the mechanical rules of the game. By April 2019, OpenAI was able to defeat Esports team OG in a full 5 versus 5 match, becoming the first AI to beat the world champions at an Esports game.

Earlier this year, Google's DeepMind developed an AI – AlphaStar, to play Starcraft II – a real time strategy game where players compete to harvest resources, build more advanced structures and develop new capabilities that can be used to destroy their enemy. Like OpenAI, DeepMind also made use of neural networks to train the bot, allowing it to beat top Starcraft pros with ease. They then also went one step further, and for a period, allowed anyone to play against AlphaStar. Not only did this help train the bot, it also improved the skill of many players worldwide.

So, what is the future of AI in gaming? Whereas earlier, some AIs (known as 'cheating AIs') retrieved opponent data from the game engine in order to stand a chance against a human, the integration of machine learning algorithms means computer-controlled enemies can compete without the need to cheat. This not only makes gameplay feel more realistic, but it can be applied in professional play, where Esports teams are potentially coached by an AI algorithm. Some people may even find playing against an AI controlled opponent more enjoyable than against people since it is more tailored, challenging and there is a lot more they can learn. However, whereas it is possible to now create NPCs that can beat real players, above all, it is important that the use of machine learning improves user experience. As new technologies such as virtual-reality and augmented-reality continue to evolve, the

boundaries between reality and the virtual world start to blur. Coupled with AI, this could lead to ground-breaking video games in the future.

References:

‘Artificial intelligence in video games’
Wikipedia. Available at:
https://en.wikipedia.org/wiki/Artificial_intelligence_in_video_games (Accessed: 29 October 2019).

Lou, H. (2017) ‘AI in Video Games: Towards a More Intelligent Game’, SITN, 28th August.
Available at:
<http://sitn.hms.harvard.edu/flash/2017/ai-video-games-toward-intelligent-game/>
(Accessed: 29 October 2019).

Kasparov, G. (2017) ‘Deep Thinking: Where Machine Intelligence Ends and Human Creativity Begins’

‘AlphaStar: Mastering the Real-Time Strategy Game StarCraft II’ DeepMind. Available at:
<https://deepmind.com/blog/article/alphastar-mastering-real-time-strategy-game-starcraft-ii>
(Accessed: 29 October 2019).

‘OpenAI Five’ OpenAI. Available at:
<https://openai.com/five/> (Accessed: 29 October 2019).

AI in Medicine

Ekaksh Lal

Artificial intelligence (AI) is an area of computer science that emphasizes the creation of intelligent machines that work and react like humans. AI is an old concept but is coming of age only in recent times. It is becoming possible due to the advancement of technology and the level of sophistication that computing and datasets have achieved. When programming a computer, an engineer often encapsulates the problem solution in terms of algorithm - a set of rules for every possible scenario. The logical next step in this process is known as AI, where a machine mimics the cognitive functions we associate with the human mind, making automated decisions. The end game is known as machine learning - providing a machine (computer) with various problems and solutions, and allowing it to infer its own set of rules based on this data, based on neural networks (the mathematical system which identifies patterns within data and images), essentially allowing it to 'think'.

Currently AI has a limited, but important purpose in medicine. One example is the clinical decision support (CDS) systems clinicians can use in their daily practice. These are even being used unconsciously; while filling a very simple form where based upon a reading a suggested dosage of medicine is suggested. Whilst relatively basic, this represents one of the links between two complex fields. The benefits of CDS are evident. Humans, including trained clinicians, are unable to process large quantities of data all at once. We do not integrate data well and tend to interpret it negatively in a stressful situation. The machines on the other hand can do this task easily as long as the rules have been clearly defined. Indeed, in some countries this sort of technology is being applied to contact tracing for the COVID outbreak.

Still, most current CDS systems are not very sophisticated in the final prediction – the clinician has a central role in approving the prediction. This is where the AI based CDS system comes in. AI has the potential to come up with its own rules by looking at data – and it can look at much more data than an expert can. For example, an AI which studies X-rays could diagnose a patient based on the data of a million other patients, while an expert would not be able to study anywhere near a million patients. This is evident now, with Google's AI determining the gender of the patient with a striking 97% chance of success, based on retina photos. This strongly suggests that the computer can find rules that even people cannot find as many believed that the anatomy of the eye was the same or similar for both genders, but this suggests otherwise.

However, a key issue with AI now can be summed up by the term 'black box' problem. For most AI systems, understanding why exactly the machine makes a recommendation is difficult to understand – consequently clinicians are less likely to interact with something they do not know much about or trust.

Case study: Deep Mind x Moorfields

Ophthalmology, the branch of medicine dealing with the diagnosis, treatment, and prevention of diseases of the eye and visual system, requires practitioners to analyse data such as images and graphs, and diagnose the patients. It is also one of the first branches that is seeing the implementation of AI. DeepMind, a London based Google subsidiary company, has partnered with the London Moorfield Eye Hospital and the University College London Institute of Ophthalmology to investigate whether AI could help improve the care and diagnosis of patients with sight threatening diseases. Sight threatening diseases can be identified via the optical coherence tomography (OCT) scans and through the utilisation of 2 neural networks, the AI developed was able to learn how to

understand the scans and identify 10 different eye diseases. Dr Pearse Keane, the senior leading doctor of the Moorfields team stated: *“The number of eye scans we’re performing is growing at a pace much faster than human experts are able to interpret them. There is a risk that this may cause delays in the diagnosis and treatment of sight-threatening diseases, which can be devastating for patients.”*, emphasising that AI must be administered in order to assist doctors and reduce delays in the diagnostic process.

Moreover, based on the diagnosis, DeepMind’s AI made its own referral with these matching the expert’s decision more than 94% of the time and zero false negatives (cases when AI missed clear indicators of disease). Furthermore, the AI is capable of maximising its use in Ophthalmology as it provides the information to professionals on how this recommendation was concluded (i.e. compared the OCT scan of the patient with the features of the disease) allowing the professionals to assess the situation; the AI can be applied to different models of OCT scanners in the future - therefore allowing it to still be used even if the hardware is upgraded where *“eye care professionals can prioritise those with the most serious eye conditions, taking us one step closer to preventing avoidable sight loss”* – Dr Pearse Keane.

DeepMind hopes to go through with the clinical trial soon, exploring how the AI could be adapted to improve patient care, and if these are successful - with the AI proving to be safe and effective - the AI could be utilised across Moorfields’ 30 hospitals and clinics across the UK.

The Future of AI in Medicine

In the future, it is likely that AI will be utilised to improve the efficiency of the overall medical system. If results for a particular scan or test could arrive quicker, there would be an early diagnosis, which would mean that the treatment could be administered promptly. As

a result, far more lives could be saved. Also, improving efficiency could reduce the costs of running a healthcare service, meaning that money could be spent elsewhere, for example, using it to increase the amount of doctors to oversee the whole process or employing more nurses for patient care.

AI especially could assist the A&E department, whereupon arrival (in scenarios that are not immediately fatal) you encounter a triage system. This system helps assess a patient's health and direct them to the correct wards. Usually, a nurse or doctor helps determine this - yet the use of AI could benefit both parties further by providing relevant information to assist in the assessment - potentially allowing the individual monitoring to notice the early symptoms of diseases. Furthermore, decision making in stressful situations may result in the incorrect course of actions, as doctors may overlook key details. In this scenario, AI could perhaps reiterate the details once again with a suggested course of actions based on its findings, easing any tension felt by clinicians.

To diagnose a disease using scans, the clinician compares the results with the features of diseases. Replicating this process with AI instead, however, would result in fewer doctors required in analysing scans, and potentially, we could get more accurate and detailed reports due to the AI being able to conclude based on far more previous data. Furthermore, AI has the potential to go beyond the distinctions that elude human observers. For example, future technology may be capable to grade cancers or instances of cardiac diseases directly, thus distinguishing the cancers that require treatment from those that may be left untreated. This prevents unnecessary wasted time in sending samples to a lab for the analysis of the cancer.

AI has great potential in countries with lower income or do not have access to many doctors (i.e. low physicians density). The

implementation of AI as a 'chatbot' could allow people to have access to the advice they need: how serious the situation is; the correct course of action (e.g. visit the A&E or seek the necessary medicine); information about the recommended dosages of a particular medicine based on the patient's physique.

Overdiagnosis and overtreatment are common in medical services, both wasting resources and the doctor's time. The utilisation of AI in this scenario would help the doctor assess the situation, allowing the right patient to get the right doctor at the right time.

The inherent goal of automation in health care, however, is far more profound: to improve the quality and safety of services. Medicine has already an established safety culture and is subsequently ready for better and safer systems. If we can demonstrate that automated systems consistently increase patient safety, then any discussion of if and when CDS should be used to replace doctors is unnecessary.

Conclusions

Since the utilisation of AI in medicine is fairly new, it may be less accurate and reliable thereby putting patients at risk. Not only does AI have to be accurate and safe, it must be created so it is up to date with new health cases. The algorithm will need to be trained, or at least constantly updated, to be able to identify new or exceptional health cases.

AI medicine currently works on a narrow range of tasks and will need close supervision for years to come. At present, they are merely "black boxes", in that doctors do not know exactly how they reach their decisions. They do, however, help to reduce much of the drudgery and error out of diagnosis.

I do not think AI is going to make human experts redundant in the near future. Rather, I believe that the two will work together, hand in hand, making machine learning algorithms

an effective partner of sorts in patient treatment. Whilst one of the founding principles of medicine may be the traditional patient-doctor relationship, the insight from a third component may make the whole process more easy, quick, and accurate.

References:

- Hannon, P., 2020. Researchers Say Use of Artificial Intelligence in Medicine Raises Ethical Questions. [online] Medicalxpress.com. Available at: <<https://medicalxpress.com/news/2018-03-artificial-intelligence-medicine-ethical.html>> [Accessed 10 July 2020].
- Khan, J., 2020. The Promise and Perils of AI Medical Care. [online] Bloomberg.com. Available at: <<https://www.bloomberg.com/news/articles/2018-08-15/the-promise-and-perils-of-ai-medical-care>> [Accessed 10 July 2020].
- Medium. 2020. The Impact of Artificial Intelligence in Healthcare. [online] Available at: <<https://medium.com/@Unfoldlabs/the-impact-of-artificial-intelligence-in-healthcare-4bc657f129f5>> [Accessed 10 July 2020].
- OECD. 2020. Using Artificial Intelligence to Help Combat COVID-19. [online] Available at: <<https://www.oecd.org/coronavirus/policy-responses/using-artificial-intelligence-to-help-combat-covid-19-ae4c5c21/>> [Accessed 10 July 2020].
- The Economist. 2018. Artificial Intelligence Will Improve Medical Treatments. [online] Available at: <<https://www.economist.com/science-and-technology/2018/06/07/artificial-intelligence-will-improve-medical-treatments>> [Accessed 10 July 2020].
- UCL News. 2020. Artificial Intelligence Equal to Experts in Detecting Eye Diseases. [online] Available at: <<https://www.ucl.ac.uk/news/2018/aug/artificial-intelligence-equal-experts-detecting-eye-diseases>> [Accessed 10 July 2020].

Tracking the Spread of Contact Tracing

Pierce Wiegerling

Among the myriad of methods being employed to stem the current spread of SARS-CoV-2, contact tracing is one of the last to be implemented, and the most difficult to get right. Other methods are straightforward, like lockdown or mask-wearing, but the line between achieving the efficacy required, while maintaining the privacy and security of citizens, is not the easiest line to follow.



Let us start by addressing the kerfuffle surrounding England's tracing app. England started with NHSX, a part of the government that develops technology for the NHS, developing a contact tracing app for public use, which had a few problems. A key reason behind the dismissal of the first iteration of the app was its focus on user-input symptoms, rather than solid testing. While it is reasonable that using these self-input symptoms could help others to isolate quicker, containing the spread of the virus better, people are famously unreliable; either by not inputting symptoms or by over-inputting ambiguous symptoms. Such an app would become a torrent of notifications, as you are constantly notified of the potential symptoms sent in by people you have been in the vicinity of in the last 14 days. Naturally, as with any such app, this would cause many to simply silence or ignore the app entirely, rendering it useless for the few times it is right. A large roadblock that contributed to this issue is the fact that testing in the UK wasn't up-to-scratch; meaning that only

notifying potential 'infectees' when someone they were in contact with was confirmed to have the virus, was much too slow of a method to authorise for widespread use, as the notifications would come far too late.

The mechanism by which the second iteration of the NHS app worked is as follows. Starting with the 'admin-work' behind the methodology of the app: when the app was downloaded, the app would immediately register a unique key with that device, and phone users always had to have their Bluetooth on. The phone would then detect when another individual's phone was within its proximity and then the two devices would exchange these unique keys with each other, noting that the owners of the devices had 'come into contact'. If within a couple of weeks, one of those people is diagnosed with CoViD-19, they would enter this (more conclusive) result in the app. The whole time, everyone's phones will periodically be downloading the keys of any, confirmed infected people within their area. If the phone recognises a key that it is downloaded, it will warn the phone owner of this and recommend that they self-isolate. Hopefully, the case where the phone recognises the key of a device that it's currently exchanging with, within its current 'blacklist', is something that won't occur, as we'd hope that those who have been tested positive wouldn't be walking around outside. Another point to address is that the method uses anonymous keys, meaning that the devices and owners can be tracked down if necessary, but their identity is not easily discernible by others using the app. The second key factor was a lack of the integration that would have come with a proprietary app (Apple and Google seeing as their OSs dominate the smartphone market), as well as a lack of discrimination in the distance between the people in question. Meaning that the app did not care how close the two individuals were if it could exchange keys, it would. But, more importantly, the lack of integration leads to issues when phones

became idle; such as, the phones not registering the keys of phones that they were in proximity to because the Bluetooth was temporarily switched off as Apple had set their devices to do when idle, in order to save power. Rendering the app completely useless during idle times.

A potential solution for the proximity issue that may have been considered by some (myself included) is the use of GPS on mobile devices and using GPS tracking to see which people were in close proximity with one another. Ignoring the blatant privacy violations and data processing that come with tracking everyone's location, all the time, there is another more inherent issue with GPS technology that discourages the use of the technology. Have you ever noticed that Google Maps sometimes like to place your location in places where it most definitely is not? For example, if you are walking down a street and Google suddenly decides that you are invading someone's home. Ignoring the cool physics that is used, a quick overview of GPS shows that it uses a technique known as trilateration to pinpoint locations of devices, whereby the distances between your phone and at least 3 satellites, are used to construct spheres of that radius and where all of those spheres intersect is likely where you are situated. While newer GPS satellites could be so accurate that they allow this method to be viable, older satellites are only accurate to a radius of several metres, making this approach unviable (for now).

The solution that is now looking to be implemented, takes advantage of the big firms that control the systems being run by most people's mobile devices. Google and Apple are working together on an API that can be added to the apps created by countries' healthcare systems to properly integrate these apps into their respective devices. One thing that I (purposefully) failed to mention above, is that the original NHS app made use of a cloud, wherein data from the app would be sent and collated by the NHS in the hopes

of using that data for extra research. This is in stark contrast to Apple and Google's API, which uses a "decentralised" model that keeps all data on the device, for the sake of the user's privacy (ironically). This new system would record contact when the devices were within a couple of metres of each other, for a few minutes; this ensures that only 'meaningful' contact is recorded. These apps would also be able to send unimpeded notifications to the user to warn them of potential exposure and prompt them to self-isolate, essentially that the notifications won't be blocked in the same way that users may disable notifications for annoying apps like the first iteration NHS app. Bluetooth isn't amazing at measuring exact distances due to the signal strength being easily affected by nearby objects, but it's good enough to roughly tell the difference between moderately close proximity and people who are further apart. Another advantage of this collaboration is the added compatibility; devices running different apps from different healthcare systems can still communicate, as most of these apps will be utilising the same API. In terms of privacy and security, these apps will be a lot better at providing that, with the only piece of data collected being the postcode area of the user, as this can be used to identify hotspots of breakouts. Some countries still wish to use GPS tracking to monitor the movement of their citizens, but Google and Apple have made it clear that, if that be the case, they wish to not be a part of it.

England is still months away from having a Contact Tracing app, with the date being postponed to winter. The government have made it clear that while it is important, the app is not one of their top priorities, but continues to assure us that it is in the works. Either way, the contribution of Apple and Google provides a huge boost to the compatibility and functionality of this app, but we will have to wait and see if the government can take advantage of it.

Python Courses – Should You Do Them?

Neil Tripathi

There are several reasons why Python is one of the most popular programming languages in the world; easily readable and very powerful, the high-level language is incredibly simple to learn yet difficult to master. Python is an extremely versatile language, and has applications ranging from simply learning the basics of programming to coding Artificial Intelligence – and is even used in many of Google’s search algorithms!

All of these reasons are why Python is the fastest growing programming language, and I could see the future potential of being proficient in Python. The best way I could think of to do this was to complete a course on Coursera – this would not only teach me Python, but also give me a certificate upon completion that I could use in a LinkedIn profile, job applications etc. I thought it would be easy, especially since I was already fluent in python. And, early on, it was.

In this article I will be sharing my experience of a Coursera Python course and whether it was worth doing. The course I completed is called ‘Python 3 Programming Specialization’ and is run by The University of Michigan, a highly prestigious university. The specialization is split into 5 sections, with each course increasing in difficulty.

Course 1 – Python Basics

For anyone fluent in Python, completing this course is rather tedious, as you have to watch all of the videos and complete all of the exercises that teach you how to print(‘Hello World’).

However, it could be useful to help touch up on the basics.

For a beginner, this course is a comprehensive introduction to Python, consisting of several videos, exercises, and interactive textbooks. The professors cover all of the basics of Python in an organised system, starting with operators, variables and function calls. There are 4 ‘weeks’ of learning materials: in total, there are 71 videos, 81 readings, 3 quizzes and 10 graded assessments.

If you are a Python beginner, it would be very hard to stay a beginner after completing the huge amounts of exercises, so I would say the course is definitely worth it. If you are already fluent, you should be able to speed through this course. Trust me – it gets harder.

Course 2 – Functions, Files and Dictionaries

This course has less content than the first but in my opinion the content is of a higher quality than course 1. It is still very easy, but it focusses on harder topics than the first such as functions, iteration, and dictionaries. I noticed that the difficulty increased considerably from the last course, requiring you to pay far more attention. There were 5 ‘weeks’ of material in this course, and I would say that it would take just as long as course 1 to complete.

However, this course thoroughly covers some of the most important topics in Python (functions and file handling), so I suggest you pay attention while doing this course. It should still be a breeze if you are fluent in Python.

Course 3 – Data Collection and Processing with Python

This course was substantially harder than the previous courses due to the fact that it dealt with APIs (application programming interface), which can be quite tricky to learn in Python. In my opinion this was one of the more important courses due to it teaching an important real-world application of Python. The first half deals with nested data, iteration and the important map() function, which are

all useful when using APIs. The second half intuitively taught you how to use APIs, but the thing I found most useful was the exercise: you had to communicate with a real website and retrieve data.

The reason I found this useful was because although there are already many tutorials on the basics of Python, I had not come across APIs previously. The topic is very complex and has a huge number of real-world applications, and therefore can be difficult to learn. However, I found it surprisingly easy to learn due to the clear, detailed instructions provided.

I would argue this was one of the most useful courses.

Course 4 – Python Classes and Inheritance

This is one of the most important topics in Python – classes are complex data structures that provide object-orientated functionality into Python, which is a very powerful way to program. This course was far shorter than the previous courses (with only 40 videos, 27 readings, 2 practice quizzes and 5 graded projects), but this is due to the difficulty of learning and understanding what classes and how to use them. Learning about OOP is very important for any aspiring programmer, so I believe this is also one of the more important courses.

This course was by far the most interesting, because OOP is such a thought-provoking paradigm of programming. I found the videos the clearest way of explaining OOP due to the professor drawing how exactly classes, objects and methods relate to each other.

Course 5 – Python Project: Pillow, Tesseract and OpenCV

This course proved to be less beneficial than the others. With only 20 videos, 5 readings and 2 graded assignments, the professors had clearly run out of time. To make things worse, this project covered some very major and difficult topics in Python that required far

more teaching than was provided: pillow (or PIL) is a powerful image-editing module in Python. Like most advanced topics in Python it is very complex and difficult to learn.

Tesseract is an optical character recognition tool in Python. That may sound complex – and it is. Luckily, it is easier to use, and the course did a better job explaining it. OpenCV is used for real-time computer vision, and in this course, it is used to detect faces in images.

The final project was very tough because I had to learn almost everything by myself. Only the task was provided with a few lines of code to start you off, and no other information was given. I struggled to combine 3 very different, rather difficult modules, and I am disappointed by the lack of guidance in this final course.

Perhaps the professors' expectations were too high, or they thought the project wasn't that difficult, or maybe they simply wanted to challenge me. Personally, I did not enjoy the final course because of how much I had to rely on information outside of the course when it was supposed to teach me.

Final Verdict

There were some downsides. The main issue was the hefty £37 per month you have to pay for access to the course. It took me just over a month to complete this specialization, and for a complete beginner I estimate it would take between 2-4 months to complete. This is why I recommend learning at least some of the basics of Python before beginning this course to be more efficient and to save money. Also,

Taking everything into account, I absolutely recommend completing this specialization. The first 4 courses have extremely high value for money, and after successfully completing them you should be very fluent in Python. Even though the final course was lacking in guidance, I still learnt a great deal just from completing the final exercise – the ecstatic rush from completing it was definitely worth it! Not only that, the certificate provided can

be used in your CV which could help you find a job. Moreover, having programming as one of your skills is becoming increasingly important, and could lead to you having a high-skilled, high-paying job. This is one of the better online Python courses, coming from a reputable university, which is why I suggest you complete this course.



5 Courses

- Python Basics
- Python Functions, Files, and Dictionaries
- Data Collection and Processing with Python
- Python Classes and Inheritance
- Python Project: pillow, tesseract, and opencv



07/01/2020

Neil Tripathi

has successfully completed the online, non-credit Specialization

Python 3 Programming

This specialization teaches the fundamentals of programming in Python 3. We will begin at the beginning, with variables, conditionals, and loops, and get to some intermediate material like keyword parameters, list comprehensions, lambda expressions, and class inheritance. You will have lots of opportunities to practice. You will also learn ways to reason about program execution, so that it is no longer mysterious and you are able to debug programs when they don't work. By the end of the specialization, you'll be writing programs that query Internet APIs for data and extract useful information from them. And you'll be able to learn to use new modules and APIs on your own by reading the documentation. That will give you a great launch toward being an independent Python programmer.

The online specialization named in this certificate may draw on material from courses taught on-campus, but the included courses are not equivalent to on-campus courses. Participation in this online specialization does not constitute enrollment at this university. This certificate does not confer a University grade, course credit or degree, and it does not verify the identity of the learner.

Stephen Oney
Steve Oney
Assistant Professor
School of Information

Paul Resnick
Paul Resnick
Michael D. Cohen
Collegiate Professor
School of Information

Jaclyn Cohen
Jaclyn Cohen
Lecturer
School of Information

Christopher Brooks
Christopher Brooks
Research Assistant
Professor
School of Information

Verify this certificate at:
coursera.org/verify/specialization/SCGH72TNJ5S8



COMPUTER SCIENCE SOCIETY

COMPUTER SCIENCE SOCIETY MEETS EVERY MONDAY IN IT2 AND IS AIMED AT GOING BEYOND THE GCSE AND A-LEVEL CURRICULUM TO ENCOURAGE INTEREST AND BROADEN KNOWLEDGE IN ALL ASPECTS OF COMPUTER SCIENCE. WE HOST TALKS AND EVENTS ON A WIDE VARIETY OF TOPICS OPEN TO ALL YEARS TO FOSTER A WIDE RANGE OF OPINIONS.

COMPUTER SCIENCE IS A DYNAMIC, RAPIDLY DEVELOPING PART TO MODERN LIFE AND HAS MANY TRANSFERABLE PROBLEM-SOLVING SKILLS THAT ENABLE YOU TO SOLVE COMPLEX CHALLENGES AND MAKE A POSITIVE DIFFERENCE IN THE WORLD. OUR AIM FOR THE SOCIETY IN THE UPCOMING YEAR IS HIGHER INCLUSIVITY - WE WANT TO MAKE SURE EVERYONE IS INVOLVED, REGARDLESS OF PRIOR KNOWLEDGE OR WHAT YEAR GROUP YOU ARE IN - THERE WILL BE SOMETHING FOR EVERYONE!



HYPERTEXT

ISSUE ONE • JULY • 2020



THANK YOU FOR READING!